Amplitude and Fourth-Moment Obstructions for Prime-Corrected Sublevel Approximations to Riemann Zeta Zeros

Víctor Casaña-Giner

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Abstract

We study the author's Euler-prime sublevel points τ_k^{C}, their MCOM-clipped version τ_k^{Cm}, and the distinct weighted compensated family $\tau_k^{\mathrm{Cms}}(X;W)$. The effective MCOM bound $|S(t)| \le 0.08382 \log t + 4.444$ for $t \ge 1$ provides a uniform enclosure for the true counting discrepancy; the author's sharper asymptotic coefficient $C_{\mathrm{H}} < 0.0748596142711$ supplies a growing margin inside the fixed MCOM truncation. We retain the earlier amplitude, fourth-moment, sublevel, numerical, correlated-displacement and local-model results, carefully separating exact identities from conclusions depending on the cited IJNT/Fujii estimates. New deductions include a localized crossing-budget inequality for the actual compensated ordinal points; a transport estimate $O(T/(\log T)^3)$ between clipped and unclipped ordinal sequences conditional on the cited fourth moment; a finite-step prime-band inclusion; an exact Fourier-box formula with endpoint terms and a parity-sensitive bound for repeated heights; and an exact Euler–pole transform yielding a sufficient weighted-L^2 criterion for RH, together with its MCOM-excess formulation. The transform criterion does not assume convergence in measure of the prime phases. It is not claimed that the required L^2 bound, arithmetic noncollapse or the weighted zero-explicit formula has been proved. The imported weighted kernel is explicitly distinguished from the author's original prime construction. The paper does not establish RH, simplicity or an improved Selberg–Fujii fourth-moment theorem.

Keywords: Riemann zeta function; Riemann–von Mangoldt formula; prime powers; Selberg approximation; sublevel measures; fourth moments; argument of zeta.
MSC 2020: 11M26, 11M06, 11N05.

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List of variables and notation

Symbols are listed by first appearance. Where the same letter is used with different meanings, the scope of each meaning is indicated; the context (subscripts, arguments) always distinguishes them.

Symbol	Meaning	Introduced
$\rho_k = \beta_k + i\gamma_k$	nontrivial zeros with $\gamma_k > 0$, with multiplicity, $\gamma_k \le \gamma_{k+1}$	§1
$N(t)$	number of zeros with $0 < \Im\rho \le t$ (right-continuous)	§1
$N_0(t)$	Riemann–von Mangoldt main term (1.1)	(1.1)
$S(t)$	$\frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$; at zero heights defined by (1.2)	(1.2)
$R(t)$	$\vartheta(t)/\pi + 1 - N_0(t)$, $\lvert R(t)\rvert \le 0.53/(\pi t)$ for $t \ge 1$	(1.2)
$\vartheta(t)$	Riemann–Siegel theta function	§1
$N_\vartheta(t)$	$\vartheta(t)/\pi + 1$	Proposition 13.3
$\mathrm{clip}_a(x)$	symmetric truncation to $[-a, a]$	(1.3)
$u_k(F)$	sublevel measure $\mathrm{meas}\{F < k - \frac{1}{2}\}$	(1.4)
k, α_F, H, Y, h	parameters of the Fujii input (moment order, range exponent, interval length, prime cutoff, shift)	(1.5)–(1.6)
p_ν, m_j	ν-th prime; prime budget $\lfloor 2^{j/2}\rfloor$ on the dyadic block $[2^j, 2^{j+1}]$	(2.1), (2.2)
Φ_m, Φ°	Euler phase of the first m primes; its dyadic interpolation	(2.1), (2.2)
$h(t)$	original truncation $\log\log\max(t, e^e)$ (in §17 the letter h also denotes Lg)	(2.3)
F_h, τ_k^{C}	original IJNT function and its sublevel points	(2.3)
$A(t)$	generic truncation bound in Lemma 2.1	Lemma 2.1
q	in Lemma 2.1: $N(t)$; elsewhere $q(v)$ is the weight density (5.1) and q_F the integer discrepancy	–
r	index displacement (Theorem 2.2); prime-power exponent ((2.1)); radii r_1, r_2 (§18)	–
$H(t)$	MCOM truncation $0.08382\log\max(t, e) + 5.5$	(3.2)
$A_{\mathrm{M}}, B_{\mathrm{M}}$	0.08382 and 5.5	Proposition 21.3
C_{H}	asymptotic MCOM coefficient 0.0748596142710907…	(3.9)
$G(t)$	$N(t) - N_0(t)$ (in §18, G, G_L also denote generic functions)	(3.2)
F^{Cm}, τ_k^{Cm}	taucm: MCOM-truncated function and its sublevel points	(3.3)
$q_F(t)$	integer discrepancy $\lvert\lfloor F + \frac{1}{2}\rfloor - N\rvert$	Theorem 3.3
F_Q, $u_k(Q)$, Q	structural saturation model and its sublevel measures; oscillation parameter	(4.1)
$\varphi_\pm$	$N_0 \pm H$	Theorem 4.1
$\mathcal{W}$, W, q	weight class, weight, density $q = -W'$ on $[1, 2]$	(5.1)
$W_{1/4}$, $q_{1/4}$	weight with $q(v) = v/2 + 1/4$ (the index labels the weight)	(5.1), (17.1)
Λ	von Mangoldt function	§5
X, L	prime cutoff; $L = \log X$	(5.2)
Φ_X^W, Π_X^W, P_X^W	weighted Euler phase, pole compensator, $P_X^W = \Phi_X^W - \Pi_X^W$	(5.2)–(5.4)
F_X^W, $\tau_k^{\mathrm{Cms}}(X; W)$	taucms: truncated weighted function and its sublevel points	(5.5)
$\mathrm{DBSRH}(W)$	$\lim_{X\to\infty}\tau_k^{\mathrm{Cms}}(X; W) = \gamma_k$ for each k	(5.6)
a_p, $D(t)$, D_T	coefficients $1 - W(\log p/\log X)$, the difference sum (6.3), $D_T = \log(\log T/\log X)$	Theorem 6.1
A_X, B_X, b_n, w_p, V_X, δ	prime and prime-square sums, coefficients of A_X^2, weights, variance, $\delta = \min(\varepsilon/2, 1/200)$	Theorem 7.1
B_W, $\mathcal{E}_{T,X}$	bound for q of the two fixed weights; saturated set	Theorem 7.3, Proposition 7.4
$\mathcal{Z}_L$	smoothed zero sum	(8.1)
$\mathcal{L}_q(L, z)$	Laplace integral, $z = d + i(\gamma - t)$, $d = \beta - \frac{1}{2}$	(9.1)
$\mathcal{A}_k^\pm$	signed sublevel contributions	(10.2), (10.3)
$\bar{u}_k$	$(u_k + u_{k+1})/2$	§10
γ_k^{RvM}, v_k	solution of $N_0 = k - \frac{1}{2}$ with $\gamma_k^{\mathrm{RvM}} > 2\pi$; $v_k = \gamma_k^{\mathrm{RvM}}$ in proofs	§12
$\Delta_k(X; W)$, s_k	consecutive spacing of taucms; $s_k = \Delta_k$	(12.2), (21.13)
a_k, b_k	barriers $N_0 \pm H = k - \frac{1}{2}$ (not the moment bounds a_T, b_T)	Theorem 12.1
$I_k^\pm$, $B_k^\pm$	occupation intervals and saturated measures	Proposition 12.3
$e_k(F)$	evaluation defect $k - \frac{1}{2} - F(\gamma_k)$	(13.1)

Symbol	Meaning	Introduced
Ψ_j, j	$\int_{z-1/2}^{z+1/2} v^{2j}dv$; moment order (in (2.2) j is the dyadic index)	(13.2)
M	number of indices in repeated-height groups	(13.5)
$\mathscr{F}_X$	generic family in the forward criterion	Proposition 13.3
θ	prime-support exponent, $X = t^\theta$ (not ϑ)	§14
g_k	in §14: normalized nearest-neighbour separation; from §16 on: $\gamma_{k+1} - \gamma_k$	(14.1), (16.1)
δ_k	$\frac{1}{2} - \lvert e_k(F)\rvert$	§14
η, $\eta_T(F)$	mean of $e_k(F)^2$ on a block (unrelated to η_k below)	§14, §15
M_T, D_F, Φ_c	$N(2T) - N(T)$; $N - F$; truncated phase of the numerical script	(15.1)
A_T, B_T	$\int D_F^2 N_0'$ and $\int D_F^2 \Phi_c'$ on $[T, 2T]$	(15.1)
a_T, b_T, κ_T, r_T	second moment, fourth moment, derivative ratio, endpoint bound	(15.3)
$\mathcal{E}_\partial(F;T)$	endpoint contribution in (15.2)	(15.2)
$d_k(X;W)$	true-zero position error $\tau_k^{\mathrm{Cms}} - \gamma_k$	(16.1)
$\mathcal{M}_k(X;W)$	symmetric margin	(16.2)
$\mathscr{L}_{k,X}$	local Lipschitz constant of F_X^W on $[a_k, b_{k+1}]$	(16.4)
$K(z)$, μ_m, $w(y)$	two-source kernel, moments of $q_{1/4}$, kernel weight $\int_{\max(1,y)}^2 q_{1/4}$	(17.1), (17.2), Theorem 17.3
$C(z,h)$, $\mathcal{R}_K$	pair profile $K(z-h/2)+K(z+h/2)$ and its remainder; $h = Lg$	Theorem 17.2
$\mathcal{C}_{c,g,L}$, c	complete local pair profile, centre	(17.7)
F_L°	untruncated $N_0 + P_{e^L}^W$	§18
$\mathscr{D}_{L;r}$, $\mathscr{Q}_{L;r_1,r_2}$, $\mathscr{B}_{r_1,r_2}$	symmetric difference, spatial filter, filter symbol	(18.1), (18.2)
B, $\mathcal{R}_B$	smooth background and its Taylor remainder	(18.4)
$\dot{P}_L$	$\partial_L P_{e^L}^W$ at fixed t	(18.5)
A, $\mathbf{v}$, P, $\mathbf{y}$, $\mathbf{b}$, $\mathbf{r}$, $\mathbf{r}_4$, $\widehat{h^2}$	sampling matrix, K'' vector, projection, data, nuisance, residual, quartic remainder, estimator	Theorem 19.1
h_k^{RvM}, $\varepsilon_k^{\mathrm{RvM}}$	RvM spacing; RvM displacement $\tau_k^{\mathrm{Cms}} - \gamma_k^{\mathrm{RvM}}$	(21.1), (21.2)
η_k, X_k	coherence fraction and threshold cutoff	(21.6)
$\xi_{k,\pm}$	mean-value points	Proposition 21.3
J_k, p, K_k	band $[a_k, b_{k+1}]$, Lebesgue exponent, uniform derivative bound	Proposition 21.7
$c_k(X;W)$	pair centre $\frac{1}{2}(\tau_k^{\mathrm{Cms}} + \tau_{k+1}^{\mathrm{Cms}})$	(21.13)
ε	fixed small exponent in the cutoff ranges (unrelated to $\varepsilon_k^{\mathrm{RvM}}$)	Theorem 7.1
$E_X(t)$	excess $(\lvert P_X^W(t)\rvert - H(t))_+$ of the *untruncated* weighted prime phase	(26.5)
u_k°	auxiliary untruncated sublevel points (not the author's original taucs)	(24.1)
$\mathcal{T}$ and k_s	auxiliary Fourier–Laplace transform of the weighted Euler phase	(26.1)
Ψ_g	finite-interval box convolution of a test function	Theorem 28.1
$J(t)$, $r_0(t)$	Dirichlet-type integral and the limit correction $r_0 = -1 + \pi^{-1}\arctan(2t) + \pi^{-1}J(t)$	Lemma 27.1
$E(s)$	$s^{-2} - 2(s-\frac{1}{2})^{-2} + \int_1^2 (\log u)u^{-s}du$, the transform of r_0	Lemma 27.1
P_∞, F_∞	limit in measure of $P_{X_j}^W$; $N_0 + P_\infty$	Proposition 27.3
$t_\sharp$	$\mathrm{meas}\{0 < t < \gamma_1 : R(t) + r_0(t) \le -\frac{1}{2}\}$, the first-index defect	Proposition 27.3
$\tau_k^\flat(X;W)$	$2\pi + \mathrm{meas}\{t \ge 2\pi : F_X^W < k - \frac{1}{2}\}$, sublevel points measured from 2π	Proposition 27.3
$\mathcal{K}_X(s)$	$\sum_k k_s(\tau_k^\flat(X;W))$	Proposition 27.5

1 Definitions and scope

Let $\rho_k = \beta_k + i\gamma_k$ denote the nontrivial zeros of $\zeta(s)$ with $\gamma_k > 0$, counted with multiplicity and indexed so that $\gamma_k \le \gamma_{k+1}$. No assumption $\beta_k = \frac{1}{2}$ is made. Let $N(t)$ count these zeros with $0 < \Im\rho \le t$, including multiplicity, and put

$$N_0(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8} \qquad (t > 0). \tag{1.1}$$

The unconditional Riemann–von Mangoldt identity, away from the imaginary parts of zeros, is

$$N(t) = N_0(t) + S(t) + R(t), \qquad S(t) = \frac{1}{\pi}\arg\zeta(\tfrac{1}{2}+it), \qquad R(t) = O(t^{-1}). \tag{1.2}$$

Here $R(t) = \vartheta(t)/\pi + 1 - N_0(t)$, with ϑ the standard Riemann–Siegel theta function. At the imaginary parts of zeros, N is right-continuous ($N(t)$ counts the zeros with $0 < \Im\rho \le t$), and $S(t)$ is *defined* there by (1.2), that is $S(t) := N(t) - N_0(t) - R(t)$; thus (1.2) holds for every $t > 0$ and S is right-continuous. Bounds for $|S|$ stated away from the zero heights extend to all $t > 0$ by taking right limits, since the bounding functions are continuous. The distinction between $N(t)$ and the count of zeros on the critical line is essential throughout.

For $a \ge 0$ and $x \in \mathbb{R}$ define the symmetric truncation

$$\mathrm{clip}_a(x) = \max\{-a, \min\{a, x\}\}. \tag{1.3}$$

For a continuous function $F : (0,\infty) \to \mathbb{R}$ whose sublevel sets below the levels $k - \frac{1}{2}$ have finite measure, put

$$u_k(F) = \mathrm{meas}\{t > 0 : F(t) < k - \tfrac{1}{2}\}, \qquad k \ge 1. \tag{1.4}$$

No monotonicity assumption on F is made. The central problem is to determine how accurately $u_k(F)$ approximates γ_k, including the possible occurrence of equal γ_k.

Status convention. An *unconditional* assertion means no RH or simplicity assumption. References to the IJNT fourth-moment and gap estimates are expressly identified as inputs of the cited manuscript, rather than independently reverified here. Statements about the full weighted Euler phase as $X \to \infty$ are *not* inferred from bounds proved only for $X \le T^{1/4}$. Section 9 lists the analytic steps that remain open. Sections 16–20 further isolate the local two-source model from the actual prime phase, including its residual; Section 21 gives the RvM-displacement comparison.

The Fujii input. All uses of Fujii's theorem below refer to the statement in [1] (Theorem "Fujii, for the modulus 1, with the corrections of [11, §5]", label `thm:known_fujii`): Fujii's Main Theorem [10, p. 225], together with Selberg's approximation as given in [10, §3, pp. 232–233] and [10, Lemma 5]. Let $\mathsf{k} \ge 1$ be an integer, let $0 < \alpha_{\mathrm{F}} \le \frac{1}{2}$ be fixed, and let $T^{1/2+\alpha_{\mathrm{F}}} \le H \le T$.

(a) For $T^{\alpha_{\mathrm{F}}/(20\mathsf{k})} \le Y \le H^{1/\mathsf{k}}$,

$$\int_T^{T+H} \Big| S(t) - \frac{1}{\pi}\,\mathrm{Im}\sum_{p<Y} p^{-1/2-it}\Big|^{2\mathsf{k}} dt \ll_{\mathsf{k},\alpha_{\mathrm{F}}} H. \tag{1.5}$$

(b) For $0 < h \le 1$ and $T \ge T_0(\mathsf{k}, \alpha_{\mathrm{F}})$,

$$\int_T^{T+H} \big(S(t+h) - S(t)\big)^{2\mathsf{k}} dt \ll_{\mathsf{k},\alpha_{\mathrm{F}}} H\big(\log(2 + h\log T)\big)^{\mathsf{k}}. \tag{1.6}$$

Only $\mathsf{k} \le 2$ and $H = T$ are used below; as explained in [1], the corrections of [11, §5] do not affect (a) or (b) in this range.

2 The original taucs and their amplitude obstruction

Write $p_1 < p_2 < \dots$ for the rational primes and set

$$\Phi_m(t) = -\frac{1}{\pi}\sum_{\nu\le m}\arg(1 - p_\nu^{-1/2-it}) = -\frac{1}{\pi}\sum_{\nu\le m}\sum_{r\ge1}\frac{\sin(rt\log p_\nu)}{r p_\nu^{r/2}}, \qquad \Phi_0 = 0. \tag{2.1}$$

For $2^j \le t \le 2^{j+1}$ define the dyadically interpolated phase

$$m_j = \lfloor 2^{j/2} \rfloor, \qquad \Phi^\circ(t) = \left(2 - \frac{t}{2^j}\right) \Phi_{m_j}(t) + \left(\frac{t}{2^j} - 1\right) \Phi_{m_{j+1}}(t). \tag{2.2}$$

The original construction in [1] is

$$h(t) = \log\log\max(t, e^e), \qquad F_h(t) = N_0(t) + \mathrm{clip}_{h(t)}(\Phi^\circ(t)), \qquad \tau_k^{\mathrm{C}} = u_k(F_h). \tag{2.3}$$

The IJNT manuscript establishes its sublevel construction and, for the indices in the asymptotic range, strict increase and that

$$\sum_{T < \gamma_k \le 2T} |\tau_k^{\mathrm{C}} - \gamma_k|^2 \ll \frac{T}{\log T}. \tag{2.4}$$

It does not prove pointwise containment between consecutive values of γ_k.

Lemma 2.1 (Necessary counting-function bound)**.** *Let $A(t) \ge 0$ be continuous. Suppose $N_0(t) \pm A(t)$ are eventually strictly increasing, $N_0(t) - A(t) \to \infty$, and $\sup_{0<s\le t}(N_0(s)+A(s)) = N_0(t) + A(t)$ for all sufficiently large t. If a continuous F satisfies $|F(t) - N_0(t)| \le A(t)$ and*

$$\gamma_{k-1} \le u_k(F) < \gamma_{k+1} \qquad \text{for every sufficiently large } k, \tag{2.5}$$

then at every sufficiently large t which is not the imaginary part of a zero,

$$|N(t) - N_0(t)| \le A(t) + \tfrac{3}{2}. \tag{2.6}$$

Proof. Put $q = N(t)$, so $\gamma_q < t < \gamma_{q+1}$ for large t. If $q - \frac{3}{2} > N_0(t) + A(t)$, the upper bound for $F(s)$ holds below $q - \frac{3}{2}$ on $(0, t]$, whence $u_{q-1}(F) \ge t$. This contradicts $u_{q-1}(F) < \gamma_q < t$. If $q + \frac{3}{2} < N_0(t) - A(t)$, the lower bound for $F(s)$ holds above $q + \frac{3}{2}$ for every $s \ge t$ by eventual monotonicity, whence $u_{q+2}(F) \le t$, contradicting $u_{q+2}(F) \ge \gamma_{q+1} > t$. □

Theorem 2.2 (Unconditional failure of eventual neighboring-zero containment)**.** *For the original τ_k^{C} of (2.3), the inequalities*

$$\gamma_{k-1} \le \tau_k^{\mathrm{C}} < \gamma_{k+1} \tag{2.7}$$

cannot hold for all sufficiently large k. For every fixed integer $r \ge 1$ there are arbitrarily large k with $\tau_k^{\mathrm{C}} \ge \gamma_{k+r}$ and arbitrarily large k with $\tau_k^{\mathrm{C}} \le \gamma_{k-r}$. The same statements hold for the sublevel points of any continuous phase correction bounded by $h(t)$, subject to the natural finiteness conditions.

Proof. The truncation h satisfies the conditions of Lemma 2.1. The unconditional Tsang $\Omega_\pm$ theorem [7, 8] gives

$$S(t) = \Omega_\pm\left(\left(\frac{\log t}{\log\log t}\right)^{1/3}\right) \tag{2.8}$$

along arbitrarily large heights away from zeros. Since $(\log t/\log\log t)^{1/3}/\log\log t \to \infty$, (2.6) with $A = h$ contradicts (1.2). For the stronger assertion, at a positive large value of $S(t)$ put $q = N(t)$ and use each index k with $N_0(t) + h(t) < k - \frac{1}{2}$ and $k + r \le q$; then $u_k(F) \ge t > \gamma_{k+r}$. The number of such indices tends to infinity along a positive Ω sequence. The negative sequence gives the analogous assertion on the other side, using $N_0 - h$. □

Corollary 2.3 (Necessary truncation scale)**.** *Under the monotonicity hypotheses of Lemma 2.1, eventual containment (2.5) is impossible if*

$$A(t) = o\Big((\log t/\log\log t)^{1/3}\Big). \tag{2.9}$$

Proof. Combine Lemma 2.1 with (2.8). □

3 Modified taucm from the explicit MCOM bound

The accompanying Mathematics of Computation submission [2] obtains, without RH,

$$|S(t)| \le 0.08382 \log t + 4.444 \qquad (t \ge 1), \tag{3.1}$$

with the compatible convention at zero heights. Its other bound is $|N(t) - \frac{t}{2\pi}\log\frac{t}{2\pi e}| \le 0.08382\log t + 5.326$ for $t \ge 1$. Combined with the explicit estimate $|R(t)| \le 0.53/(\pi t)$ for $t \ge 1$ and the elementary range $0 < t < 1$, (3.1) gives

$$|G(t)| \le H(t), \qquad G(t) := N(t) - N_0(t), \qquad H(t) := 0.08382 \log\max(t, e) + 5.5 \tag{3.2}$$

for almost every $t > 0$ (and under the stated conventions also at the jumps). Indeed $4.444 + 0.53/\pi < 5.5$; for $t < 1$ use $N(t) = 0$ and the elementary bound on $N_0(t)$.

Definition 3.1 (Improved taucm)**.** Retain the original Φ° and its prime budget $m_j = \lfloor 2^{j/2} \rfloor$, and set

$$F^{\mathrm{Cm}}(t) = N_0(t) + \mathrm{clip}_{H(t)}(\Phi^\circ(t)), \qquad \tau_k^{\mathrm{Cm}} = u_k(F^{\mathrm{Cm}}). \tag{3.3}$$

Proposition 3.2 (Contractivity of covered truncation)**.** *If $|y| \le a$, then for every real x,*

$$|\,\mathrm{clip}_a(x) - y| \le |x - y|. \tag{3.4}$$

Consequently, for almost all $t > 0$,

$$|F^{\mathrm{Cm}}(t) - N(t)| \le |N_0(t) + \Phi^\circ(t) - N(t)|. \tag{3.5}$$

Proof. Truncation to $[-a, a]$ is the nearest-point projection onto a closed interval. Apply its elementary distance inequality with $x = \Phi^\circ(t)$ and $y = G(t)$. □

Theorem 3.3 (Fourth moment and mean-square localization for taucm)**.** *Assume the following unconditional inputs as established in the cited IJNT manuscript [1]: its fourth-moment estimate for the* untruncated *phase (Lemma "Fourth moment of the counting error", label* `lem:fourth`*),*

$$\int_U^{2U} |N_0(t) + \Phi^\circ(t) - N(t)|^4 \,\mathrm{d}t \ll U \quad (U = 2^j \text{ sufficiently large}), \tag{3.6}$$

and its third-moment estimate for the successive differences $\gamma_{k+1} - \gamma_k$ (Lemma "Third moment of the gaps", label `lem:gaps`*) and deterministic sublevel-transfer theorem (Theorem "Sublevel transfer", label* `thm:transfer`*, with constant $2 \cdot 24^{1/3}$). Then, without RH,*

$$\int_T^{2T} |F^{\mathrm{Cm}}(t) - N(t)|^4 \,\mathrm{d}t \ll T, \tag{3.7}$$

$$\sum_{T < \gamma_k \le 2T} |\tau_k^{\mathrm{Cm}} - \gamma_k|^2 \ll \frac{T}{\log T}. \tag{3.8}$$

The constants are absolute and the prime budget is unchanged.

Proof. By Proposition 3.2, (3.6) passes directly to F^{Cm}. A general $[T, 2T]$ is covered by a bounded number of dyadic intervals of comparable length, proving (3.7). For large T, (3.2) and $H(t) = O(\log(2 + t))$ ensure that all sublevel sets for indices with $T < \gamma_k \le 2T$ contain $(0, T/2]$ and are contained in $(0, 3T)$, exactly as in the corresponding localization step of IJNT: the separations in N_0 across these endpoints have order $T \log T$, whereas both the correction and the difference $N - N_0$ are $O(\log T)$. Explicitly, let $T < \gamma_k \le 2T$ with T large, so that $N(T) \le k - 1$ and $N(2T) \ge k$. If $0 < t \le T/2$, then $F^{\mathrm{Cm}}(t) \le N_0(T/2) + H(T/2) < N_0(T) - H(T) \le N(T) \le k - 1$, using that $N_0 + H$ is eventually increasing and bounded on the initial range, and (3.2). If

$t \geq 3T$, then $F^{\mathrm{Cm}}(t) \geq N_0(3T) - H(3T) > N_0(2T) + H(2T) \geq N(2T) \geq k$. The deterministic sublevel-transfer inequality of IJNT, with integer discrepancy $q_F(t) = |\lfloor F(t) + \frac{1}{2} \rfloor - N(t)|$, gives

$$\sum_{T<\gamma_k\leq 2T} |\tau_k^{\mathrm{Cm}} - \gamma_k|^2 \ll \left(\sum_{\gamma_j \in [T/2, 3T]} (\gamma_{j+1} - \gamma_j)^3 \right)^{1/2} \left(\int_{T/2}^{3T} q_{F^{\mathrm{Cm}}}(t)^4 \, \mathrm{d}t \right)^{1/2}.$$

The first factor is $O(T^{1/2}/\log T)$ by the cited gap input; the second is $O(T^{1/2})$ because $q_F \leq |F - N| + \frac{1}{2}$ and (3.7) applies on a bounded collection of dyadic intervals. Their product proves (3.8). □

Remark 3.4 (Projection does not order the two truncations). The contractivity estimate (3.5) compares the improved function with the *untruncated* phase $N_0 + \Phi^\circ$; it does not yield $|F^{\mathrm{Cm}} - N| \leq |F_h - N|$. A larger truncation interval need not improve pointwise distance to a target lying outside the smaller interval. Hence the input required above is specifically the untruncated fourth moment (3.6), not a fourth-moment estimate deduced solely for F_h. The elementary bound $|F^{\mathrm{Cm}} - N| \leq 2H$ holds whenever $|N - N_0| \leq H$.

Remark 3.5 (Asymptotic refinement from the MCOM result). The same MCOM submission also states the stronger asymptotic inequality

$$\limsup_{t\to\infty} \frac{|S(t)|}{\log t} < 0.0748596142711. \tag{3.9}$$

Consequently, for every fixed $c > 0.0748596142711$ there exists T_c such that $|N(t) - N_0(t)| \leq c \log t$ for all sufficiently large t away from zero heights. One may therefore replace H by a positive continuous truncation bound that agrees with $c \log t$ above T_c, and that dominates $|N - N_0|$ below T_c. The same projection argument applies. Unlike (3.2), this refinement does not provide an effective value of T_c and cannot be used as an explicit estimate at a specified height.

Remark 3.6 (Exact scope). The modification removes the specific impossibility mechanism in Theorem 2.2: its truncation is large enough to contain $N - N_0$. It does not prove that every τ_k^{Cm} lies between the imaginary parts of neighboring zeros, nor does it establish RH or simplicity. The argument transfers established IJNT estimates to a different truncation; it does not derive a uniform result for arbitrary prime cutoffs X.

4 The structural saturation obstruction

The following is the precise mathematical content of the objection that ordering, truncation and the general sublevel-transfer inequality cannot by themselves prohibit saturation. It is not a theorem asserting that the *actual* weighted Euler phases saturate.

Theorem 4.1 (Structural saturation model). *Let H be given by (3.2). For $Q > 0$ define*

$$F_Q(t) = N_0(t) + H(t) \tanh(Q \sin(Qt)), \qquad u_k(Q) = u_k(F_Q). \tag{4.1}$$

For all sufficiently large Q, F_Q is continuous; its sublevel measures are finite, positive and strictly increasing in k; $|F_Q - N_0| \leq H$; and every deterministic sublevel-transfer inequality applicable to continuous F applies to F_Q. For each fixed sufficiently large k,

$$\lim_{Q\to\infty} u_k(Q) = \frac{1}{2} \operatorname{meas}\{N_0 + H < k - \tfrac{1}{2}\} + \frac{1}{2} \operatorname{meas}\{N_0 - H < k - \tfrac{1}{2}\}. \tag{4.2}$$

Writing $\varphi_\pm = N_0 \pm H$, which are strictly increasing near the relevant levels for large k, one also has

$$\lim_{Q\to\infty}(u_{k+1}(Q)-u_k(Q)) = \frac{1}{2}[\varphi_+^{-1}(k+\tfrac12)-\varphi_+^{-1}(k-\tfrac12)+\varphi_-^{-1}(k+\tfrac12)-\varphi_-^{-1}(k-\tfrac12)]. \quad (4.3)$$

These properties do not characterize the prime-corrected functions F_X^W and do not provide information on β_k.

Proof. The bounds and continuity follow immediately from (4.1). Since $N_0 - H \to +\infty$, all relevant sublevel measures are finite. For Q large, a point near $3\pi/(2Q)$ has $F_Q < 1/2$, whereas $F_Q(t) \to +\infty$ as $t \to \infty$. Continuity thus implies that F_Q takes every value in each interval $[k-\frac12, k+\frac12)$ on a set of positive measure, giving $0 < u_k(Q) < u_{k+1}(Q)$.

Fix a compact interval I and $c \in \mathbb{R}$ such that the level sets $\{N_0 \pm H = c\} \cap I$ have measure zero. Put $\eta_Q = Q^{-1/2}$ and $I_Q^\pm = \{t \in I : \pm\sin(Qt) \ge \eta_Q\}$. Then $\mathrm{meas}(I \setminus (I_Q^+ \cup I_Q^-)) \to 0$, and on $I_Q^\pm$ one has $|F_Q - (N_0 \pm H)| \le H(1-\tanh\sqrt{Q}) \to 0$ uniformly. Since $\mathrm{meas}(\{|N_0 \pm H - c| \le \epsilon\} \cap I) \to 0$ as $\epsilon \to 0$, it follows that $\mathrm{meas}(I_Q^\pm \cap \{F_Q < c\}) - \mathrm{meas}(I_Q^\pm \cap \{N_0 \pm H < c\}) \to 0$. Finally, for every measurable $A \subset I$, $\mathrm{meas}(A \cap I_Q^\pm) \to \frac12 \mathrm{meas}(A)$: the indicator of $\{\pm\sin(Qt) \ge \eta_Q\}$ differs from that of $\{\pm\sin(Qt) > 0\}$ on a set of measure $o(1)$, and the latter is $\frac12$ plus a $2\pi/Q$-periodic function of mean zero, which tends weakly to 0 by the Riemann–Lebesgue lemma. The level boundaries of $N_0 \pm H$ are finite in any compact interval, for sufficiently large k, because the functions eventually increase. Since $N_0 - H \to \infty$, the sublevel sets lie in a common compact interval for each fixed k. This proves (4.2). Taking differences yields (4.3); for large k the two sublevel measures equal the corresponding inverse-function values. □

Theorem 4.2 (Fourth-moment obstruction to saturation)**.** *Let $G = N - N_0$. For every fixed $T \ge 2$, the functions in (4.1) satisfy*

$$\lim_{Q\to\infty}\int_T^{2T} |F_Q(t) - N(t)|^4\,\mathrm{d}t = \int_T^{2T} (H^4 + 6H^2G^2 + G^4)\,\mathrm{d}t \ge \int_T^{2T} H(t)^4\,\mathrm{d}t. \quad (4.4)$$

Consequently no estimate $\int_T^{2T} |F_Q - N|^4\,\mathrm{d}t \le CT$ can hold for all sufficiently large T and all $Q > 0$ with an absolute C, because

$$\int_T^{2T} H(t)^4\,\mathrm{d}t \asymp T(\log T)^4. \quad (4.5)$$

This does not contradict the fixed-budget IJNT estimate: F_Q is an abstract test construction, not its Euler phase.

Proof. For fixed T, G and H are bounded on $[T, 2T]$. Weak equidistribution of the two signs, established in the proof of Theorem 4.1 and applied on each interval between consecutive jumps of N, gives the first equality; these jumps form a finite set. The integrand's average is $\frac12(H-G)^4 + \frac12(-H-G)^4 = H^4 + 6H^2G^2 + G^4$. The last estimate follows from $H(t) = 0.08382\log t + O(1)$ throughout $[T, 2T]$. □

Corollary 4.3 (A conditional exclusion principle)**.** *Suppose a family of* actual *weighted prime phases F_X^W satisfies, along some $X_n \to \infty$, the saturated two-sign distribution on every compact interval, in the precise weak-integral sense used in Theorem 4.2. Then the global estimate*

$$\sup_{X \ge T^\varepsilon}\int_T^{2T} |F_X^W - N|^4\,\mathrm{d}t \ll_\varepsilon T \quad (4.6)$$

for fixed $\varepsilon > 0$, uniformly in T, is impossible. Conversely, a proof of (4.6) would exclude any scenario for the prime phase for which this saturation assertion is independently established.

Proof. For fixed T all sufficiently large X_n satisfy $X_n \geq T^{\varepsilon}$. Take $n \to \infty$ in the asserted uniform estimate and use (4.4); then let $T \to \infty$. The resulting $T(\log T)^4 \ll_{\varepsilon} T$ is impossible. □

Remark 4.4 (What the objection does not prove). The preceding theorems show that the three elementary properties (increasing sublevel points, a bounded correction, and a universal transfer inequality) do not rule out saturation. They do *not* prove that RH is false, that an arithmetic family necessarily saturates when RH is false, or that no further theorem using the specific prime coefficients can lead to RH. Establishing an off-critical-line saturation statement requires a separate explicit-formula and non-cancellation argument; its present status in [3] is a proof sketch under additional hypotheses.

5 Smoothed prime powers and taucm(s)

Let $\mathcal{W}$ denote the class of weights

$$W(u) = \int_1^2 q(v)\mathbf{1}_{\{u \leq v\}}\, dv, \quad q \geq 0, \quad q \in BV[1,2], \quad \int_1^2 q(v)\, dv = 1. \tag{5.1}$$

Thus $0 \leq W \leq 1$, $W = 1$ for $0 \leq u \leq 1$, $W = 0$ for $u \geq 2$, and W is nonincreasing. The Selberg weight corresponds to $q(v) = 1$; the tested weight $W_{1/4}$ corresponds to $q(v) = v/2 + 1/4$. Let Λ denote von Mangoldt's function. For $X \geq 2$, set

$$\Phi_X^W(t) = \frac{1}{\pi} \operatorname{Im} \sum_{n \geq 2} W\left(\frac{\log n}{\log X}\right) \frac{\Lambda(n)}{n^{1/2+it} \log n}, \tag{5.2}$$

$$\Pi_X^W(t) = \frac{1}{\pi} \operatorname{Im} \int_2^{\infty} W\left(\frac{\log u}{\log X}\right) \frac{u^{-1/2-it}}{\log u}\, du, \tag{5.3}$$

$$P_X^W(t) = \Phi_X^W(t) - \Pi_X^W(t), \tag{5.4}$$

$$F_X^W(t) = N_0(t) + \operatorname{clip}_{H(t)}(P_X^W(t)), \qquad \tau_k^{\mathrm{Cms}}(X;W) = u_k(F_X^W). \tag{5.5}$$

The sum and the integral are finite because W is supported on $[0,2]$. The compensator Π_X^W corresponds to the pole contribution. These definitions do not use β_k.

Definition 5.1 (DBSRH for a weight)**.** For a fixed $W \in \mathcal{W}$, the convergence statement is

$$\mathrm{DBSRH}(W): \qquad \lim_{X \to \infty} \tau_k^{\mathrm{Cms}}(X;W) = \gamma_k \quad \text{for each fixed } k \geq 1. \tag{5.6}$$

No equivalence between (5.6) and RH is assumed or proved here.

Lemma 5.2 (Pole compensator)**.** *Uniformly over $W \in \mathcal{W}$, $X \geq 2$, and $t \geq 1$,*

$$|\Pi_X^W(t)| \ll \frac{1+X}{t}. \tag{5.7}$$

Proof. With $L = \log X$ and $v = \log u$, the integral in (5.3) has amplitude $a(v) = W(v/L)e^{v/2}/v$, supported on $[\log 2, 2L]$. The total variation of a plus its endpoint values is $O(1+X)$, uniformly in W: W is nonincreasing with total variation at most one, and $e^{v/2}/v$ has bounded variation $O(1+X)$ on this interval. Integration by parts against e^{-itv} proves the result. □

6 A rigorous restriction on small prime cutoffs

The next theorem demonstrates that a fourth-moment estimate uniform over *all* diverging cutoffs $X = X(T)$ is false, even without any hypothesis on RH.

Theorem 6.1 (Subpolynomial cutoff obstruction)**.** *Let $X = X(T) \to \infty$ and $\log X = o(\log T)$. For every choice $W = W_T \in \mathcal{W}$ (allowed to depend on T),*

$$\int_T^{2T} |F_X^W(t) - N(t)|^4 \, \mathrm{d}t \geq \left(\frac{1}{4\pi^4} + o(1)\right) T \left(\log \frac{\log T}{\log X}\right)^2. \tag{6.1}$$

The $o(1)$ is uniform over the indicated weights and, along the stated cutoff sequence, tends to zero. In particular, the left-hand side divided by T diverges.

Proof. Put $Y = T^{1/200}$. By (1.5) with $\mathsf{k} = 2$, $\alpha_{\mathrm{F}} = \frac{1}{5}$ and $H = T$ (so that $Y = T^{\alpha_{\mathrm{F}}/(20\mathsf{k})}$), and Hölder's inequality, its L^2 consequence is

$$\left\| S(t) - \frac{1}{\pi} \operatorname{Im} \sum_{p<Y} p^{-1/2-it} \right\|_{L^2[T,2T]} \ll \sqrt{T}. \tag{6.2}$$

For large T, $X^2 < Y$. The difference between the short prime sum and the prime terms of Φ_X^W is the imaginary part divided by π of

$$D(t) = \sum_{p<Y} \frac{a_p}{\sqrt{p}} p^{-it}, \qquad a_p = 1 - W\left(\frac{\log p}{\log X}\right), \tag{6.3}$$

where $a_p = 0$ for $p \leq X$, $0 \leq a_p \leq 1$ for $X < p \leq X^2$, and $a_p = 1$ for $X^2 < p < Y$. Mertens' theorem [13] yields, uniformly in W,

$$\sum_{p<Y} \frac{a_p^2}{p} = \log \frac{\log T}{\log X} + O(1). \tag{6.4}$$

Indeed the interval $(X, X^2]$ contributes at most $\log 2 + o(1)$, and the interval (X^2, Y) gives the claimed logarithm up to an absolute constant.

Montgomery–Vaughan [12] gives

$$\int_T^{2T} |D(t)|^2 \, \mathrm{d}t = T \sum_{p<Y} \frac{a_p^2}{p} + O(Y).$$

Also $\int_T^{2T} D(t)^2 \, \mathrm{d}t = O(Y)$ by direct integration of $(pq)^{-it}$, since $\log(pq) \geq \log 4$. Hence

$$\int_T^{2T} \left| \frac{\operatorname{Im} D(t)}{\pi} \right|^2 \mathrm{d}t = \frac{T}{2\pi^2} \left(\log \frac{\log T}{\log X} + O(1) \right) + O(Y). \tag{6.5}$$

The sum over prime powers p^r, $r \geq 2$, in (5.2) has $L^2[T, 2T]$ norm $O(\sqrt{T})$ uniformly for $X = T^{o(1)}$: for $r = 2$ use Montgomery–Vaughan on $\sum_{p \leq X} W(2 \log p / \log X) p^{-1-2it}$, whose diagonal $\sum_p p^{-2}$ converges and whose remainder is $O(\pi(X)) = o(T)$; the terms with $r \geq 3$ converge absolutely. By Lemma 5.2, $\|\Pi_X^W\|_2 \ll X/\sqrt{T} = o(\sqrt{T})$; also $\|R\|_2 \ll T^{-1/2}$. Combining these estimates with (6.2) and (6.5), and writing $D_T = \log(\log T / \log X) \to \infty$, gives

$$\|N_0 + P_X^W - N\|_2 \geq \left(\frac{1}{\sqrt{2}\pi} + o(1) \right) \sqrt{T D_T}. \tag{6.6}$$

We now check that truncation does not alter the principal size. Because $X^4 = T^{o(1)}$, a second application of Montgomery–Vaughan to the square of the prime polynomial yields

$$\int_T^{2T} |P_X^W(t)|^4 \, \mathrm{d}t \ll T \big(1 + \log\log(3X)\big)^2. \tag{6.7}$$

Indeed, the square of the prime part $\sum_{p\le X^2} W(\log p/\log X)p^{-1/2-it}$ is $\sum_{n\le X^4} c_n n^{-it}$ with $\sum_n |c_n|^2 \le 2\big(\sum_{p\le X^2} p^{-1}\big)^2 \ll (1+\log\log(3X))^2$ and $\sum_n n|c_n|^2 \le 2\pi(X^2)^2 \le X^4 = T^{o(1)}$, so Montgomery–Vaughan gives the bound for this part. The contributions of the prime squares, higher powers and pole term are handled as above. For every $a>0$, $|x-\mathrm{clip}_a(x)|^2 \le |x|^4/a^2$; therefore $H(t) \gg \log T$ on $[T,2T]$ implies

$$\int_T^{2T} |P_X^W - \mathrm{clip}_H(P_X^W)|^2\,\mathrm{d}t \ll T\frac{(1+\log\log(3X))^2}{(\log T)^2} = o(T).$$

Thus (6.6) remains valid with $N_0+P_X^W$ replaced by F_X^W. Finally, Hölder's inequality gives

$$\int_T^{2T} |F_X^W - N|^4\,\mathrm{d}t \ge T^{-1}\left(\int_T^{2T} |F_X^W - N|^2\,\mathrm{d}t\right)^2,$$

which proves (6.1). □

Remark 6.2 (Meaning of the lower restriction). A universal $O(T)$ fourth-moment estimate valid for *every* diverging cutoff is therefore impossible. The theorem forces $\liminf_{T\to\infty} \log X(T)/\log T > 0$ along any family satisfying a uniform $O(T)$ bound. It does not assert that $X \ge T^{\varepsilon}$ is sufficient when there is no upper bound on X.

7 Uniform fourth moment in a polynomial cutoff range

Theorem 7.1 (Uniform weighted fourth moment). *Fix $0<\varepsilon<\frac14$. Subject to the Fujii input (1.5), there is $C_\varepsilon>0$, independent of T, X, and W, such that for T sufficiently large,*

$$\boxed{\sup_{\substack{W\in\mathcal{W}\\ T^{\varepsilon}\le X\le T^{1/4}}} \int_T^{2T} |F_X^W(t) - N(t)|^4\,\mathrm{d}t \le C_\varepsilon T.} \tag{7.1}$$

No RH assumption is used.

Proof. Fix $\delta = \min(\varepsilon/2, 1/200) > 0$ and set $Y = T^{\delta}$. By (1.5) with $\mathsf{k}=2$, $\alpha_{\mathrm{F}} = 40\delta \le \frac15$ and $H = T$ (so that $Y = T^{\alpha_{\mathrm{F}}/(20\mathsf{k})} \le H^{1/2}$),

$$\int_T^{2T} \left| S(t) - \frac{1}{\pi}\,\mathrm{Im}\sum_{p<Y} p^{-1/2-it}\right|^4 \mathrm{d}t \ll_\varepsilon T. \tag{7.2}$$

Because $Y<X$, the difference between the prime terms of Φ_X^W and the sum in (7.2) is

$$A_X(t) = \sum_{Y\le p\le X^2} \frac{w_p}{\sqrt{p}} p^{-it}, \qquad w_p = W(\log p/\log X) \in [0,1].$$

Mertens' theorem and $T^{\varepsilon} \le X \le T^{1/4}$ give

$$V_X := \sum_{Y\le p\le X^2} \frac{w_p^2}{p} \ll \log\log(X^2) - \log\log Y + O(1) \ll_\varepsilon 1. \tag{7.3}$$

Write $A_X^2 = \sum_{n\le X^4} b_n n^{-it}$. If $d(n)$ counts ordered decompositions $n=pq$ with primes $Y\le p,q\le X^2$, then $d(n)\le 2$ and

$$\sum_n |b_n|^2 \le 2V_X^2 \ll_\varepsilon 1, \quad \sum_n n|b_n|^2 \le 2\pi(X^2)^2 \ll X^4 \le T.$$

Consequently the Montgomery–Vaughan mean-value theorem applied to A_X^2 yields

$$\int_T^{2T} |A_X(t)|^4 \, \mathrm{d}t \ll_\varepsilon T. \tag{7.4}$$

The prime-square terms are, up to the coefficient $1/(2\pi)$,

$$B_X(t) = \sum_{p \le X} \frac{W(2\log p/\log X)}{p} \, p^{-2it}.$$

Applying the same argument to B_X^2, now with frequencies $(pq)^2$, gives

$$\int_T^{2T} |B_X(t)|^4 \, \mathrm{d}t \ll T; \tag{7.5}$$

indeed $\sum_p p^{-2} < \infty$ controls the diagonal and the Montgomery–Vaughan remainder is $O(\pi(X)^2) = O(X^2)$. The sum of the prime-power terms with exponent $r \ge 3$ is bounded absolutely independently of X. By Lemma 5.2, $|\Pi_X^W(t)| \ll X/T \ll T^{-3/4}$ on $[T, 2T]$. Combining these estimates with (7.2) by Minkowski,

$$\int_T^{2T} |N_0(t) + P_X^W(t) - N(t)|^4 \, \mathrm{d}t \ll_\varepsilon T. \tag{7.6}$$

Finally, $|G(t)| \le H(t)$ by (3.2). The contractivity in Proposition 3.2, with $x = P_X^W(t)$ and $y = G(t)$, transfers (7.6) directly to (7.1). □

Remark 7.2 (The precise upper restriction). The exponent $1/4$ comes from the support $n \le X^2$ in the weighted prime polynomial and the support $n \le X^4$ after squaring it. Montgomery–Vaughan controls the remainder by $O(X^4)$. This is a limitation of this *particular fourth-moment proof*; it is not a theorem that a larger X is impossible or incompatible with RH. In particular, Theorem 7.1 does not establish (4.6).

Theorem 7.3 (Logarithmic improvement for the two fixed weights)**.** *Fix $0 < \varepsilon < 1/4$ and choose either the Selberg weight $q(v) = 1$ or $W_{1/4}$ with $q(v) = v/2 + 1/4$. Under the same Fujii input as Theorem 7.1, there exists $C_{\varepsilon,W}$ such that*

$$\sup_{T^\varepsilon \le X \le T^{1/4}(\log T)^{3/2}} \int_T^{2T} |F_X^W(t) - N(t)|^4 \, \mathrm{d}t \le C_{\varepsilon,W} T. \tag{7.7}$$

This is an extension in the weighted support parameter *X, not a claim for every $W \in \mathcal{W}$.*

Proof. For the indicated weights, $0 \le q(v) \le B_W$ and $W(v) \le B_W(2 - v)$ for $1 \le v \le 2$. Thus

$$\sum_{p \le X^2} W(\log p/\log X)^2 \ll_W \frac{X}{\log X} + \frac{X^2}{(\log X)^3} \ll_W \frac{X^2}{(\log X)^3}. \tag{7.8}$$

Indeed, for $p \le X$, use the prime-counting bound; for $X < p \le X^2$, split into strips $X^2 e^{-(r+1)} < p \le X^2 e^{-r}$ $(r \ge 0)$, use $W \le B_W(r+1)/\log X$ and $\pi(y) \ll y/\log y$ when $y \ge X$, then sum the absolutely convergent series $\sum_{r \ge 0} e^{-r}(r+1)^2$. For A_X as in the proof of Theorem 7.1, the bound $\sum_n |b_n|^2 \ll_\varepsilon 1$ is unchanged, while

$$\sum_n n|b_n|^2 \le 2 \left(\sum_{Y \le p \le X^2} w_p^2 \right)^2 \ll_W \frac{X^4}{(\log X)^6} \ll_\varepsilon T$$

throughout the enlarged range. The treatment of prime squares, higher prime powers and the pole compensator remains valid: $X^2 \ll T$ and $X/T \ll 1$. Apply Montgomery–Vaughan and the projection contraction as before. □

Proposition 7.4 (Small proportion of saturated values)**.** *Under the assumptions of Theorem 7.3, let* $\mathcal{E}_{T,X} = \{t \in [T, 2T] : |P_X^W(t)| \geq H(t)\}$*. Then*

$$\sup_{T^\varepsilon \leq X \leq T^{1/4}(\log T)^{3/2}} \operatorname{meas}(\mathcal{E}_{T,X}) \ll_{\varepsilon,W} \frac{T}{(\log T)^4}. \tag{7.9}$$

Proof. Split $\mathcal{E}_{T,X}$ according to $|G(t)| \leq H(t)/2$ and its complement, where $G = N - N_0$. In the first case $|F_X^W - N| \geq H/2$; Markov's inequality and (7.7) give $O_{\varepsilon,W}(T/(\log T)^4)$. In the second case, $|S(t)| \gg \log T$ for large T because $R(t) = O(T^{-1})$. The eighth-moment estimate of Selberg [9, 4] gives $O(T(\log\log T)^4/(\log T)^8)$ for the second set. □

Remark 7.5. The last proposition concerns a jointly growing X and T in a specified range. It provides no information about $X \to \infty$ with T held fixed.

8 A zero-sum fourth-moment lemma without an upper cutoff

A distinct estimate gives supporting evidence for considering $X \geq T^\varepsilon$ without an upper restriction, but is not an estimate for the full arithmetic phase.

Theorem 8.1 (Unconditional smoothed zero-sum bound)**.** *For* $L > 0$ *let*

$$\mathcal{Z}_L(t) = \sum_{\gamma_k > 0} \frac{1}{(1 + L|t - \gamma_k|)^2}, \tag{8.1}$$

where the imaginary parts are counted with multiplicity. For each fixed $\varepsilon > 0$*,*

$$\boxed{\sup_{L \geq \varepsilon \log T} \int_T^{2T} \mathcal{Z}_L(t)^4 \, \mathrm{d}t \ll_\varepsilon T.} \tag{8.2}$$

No RH hypothesis is required.

Proof. For $v \in \mathbb{R}$,

$$(1 + L|v|)^{-2} = \int_{|v|}^\infty \frac{2L}{(1 + Lu)^3} \, du. \tag{8.3}$$

Define $N(y) = 0$ for $y \leq 0$. By Tonelli and (8.3),

$$\mathcal{Z}_L(t) = \int_0^\infty \frac{2L}{(1 + Lu)^3} \big(N(t + u) - N(t - u)\big) \, du \tag{8.4}$$

for almost every t. On $0 < u \leq 1$, (1.6) with $\mathsf{k} = 2$, $\alpha_{\mathrm{F}} = \frac{1}{2}$ and $H = T$, together with (1.2), implies

$$\|N(t + u) - N(t - u)\|_{L^4[T,2T]} \ll T^{1/4}(1 + u \log T). \tag{8.5}$$

To see this, split the difference into the two increments of length u, shift the integration variable for one of them, and use the bound $\int |S(t + u) - S(t)|^4 \, \mathrm{d}t \ll T(\log(2 + u \log T))^2$; the main term changes by $O(u \log T)$, and $\sqrt{\log(2 + v)} \ll 1 + v$. For $u > 1$, the Riemann–von Mangoldt formula and $S(y) = O(\log(2 + y))$ yield the elementary pointwise bound

$$N(t + u) - N(t - u) \ll (u + 1) \log(2T + u), \qquad T \leq t \leq 2T. \tag{8.6}$$

Apply Minkowski to (8.4), using (8.5) on $(0, 1]$ and (8.6) on $(1, \infty)$:

$$\|\mathcal{Z}_L\|_4 \ll T^{1/4} \left(1 + \frac{\log T}{L}\right) + T^{1/4} \, O\left(\frac{1 + \log T}{L^2}\right).$$

Here $\int_0^\infty 2L(1 + Lu)^{-3} \, du = 1$ and $\int_0^\infty 2Lu(1 + Lu)^{-3} \, du = 1/L$; the tail integral is $O((1 + \log T)/L^2)$ for $L \geq 1$. When $L \geq \varepsilon \log T$ the right-hand side is $O_\varepsilon(T^{1/4})$, proving (8.2). □

Remark 8.2 (No hidden assumption about β_k). The sum in (8.1) uses only γ_k and their multiplicities. Thus (8.2) is compatible both with RH and with the existence of zeros outside $\Re s = 1/2$. To transfer it to the actual weighted Euler phase under RH, one must *first prove* an explicit formula with a kernel satisfying $|K_W(\frac{1}{2} + i\gamma, t; X)| \ll_W (1 + |t-\gamma| \log X)^{-2}$ and a suitably controlled remainder. That transfer is not contained in Theorem 8.1.

9 Off-critical zeros, weight smoothness, and the unresolved range

For W as in (5.1), the leading contribution of a hypothetical zero with $\rho = \frac{1}{2} + d + i\gamma$ $(d > 0)$, in the proposed explicit-formula analysis of [3], involves a Laplace integral of the form

$$\mathcal{L}_q(L, z) = \int_1^2 q(v) e^{vLz}\, dv, \qquad z = d + i(\gamma - t), \quad L = \log X. \tag{9.1}$$

For a sufficiently regular q with $q(2) \neq 0$, integration by parts gives, at fixed $z \neq 0$ with $\Re z > 0$,

$$\mathcal{L}_q(L, z) = \frac{q(2)e^{2Lz}}{Lz} + O_z\left(\frac{e^{2Ld}}{L^2}\right) + O_z\left(\frac{e^{Ld}}{L}\right), \tag{9.2}$$

where the displayed remainder assumes, for instance, that q has a bounded derivative. If q vanishes to finite order at the endpoint, repeated integration by parts introduces additional powers of $1/L$ but does not suppress the factor e^{2Ld}. The Selberg weight and $W_{1/4}$ have $q(2) > 0$. These statements explain a *possible* saturation mechanism, not a proved lower bound for the total zero sum: distinct off-critical contributions can cancel.

The precise remaining statement. For some fixed $\varepsilon > 0$ and a fixed admissible weight W, the candidate estimate is

$$\boxed{\sup_{X \geq T^\varepsilon} \frac{1}{T} \int_T^{2T} |F_X^W(t) - N(t)|^4 \, \mathrm{d}t \leq C_\varepsilon \qquad (T \geq T_0).} \tag{9.3}$$

The lower restriction is consistent with the unconditional obstruction in Theorem 6.1; there is deliberately no upper restriction. However (9.3) has *not* been proved. It cannot follow just from the deterministic truncation inequality, because the latter bounds the truncated error by the *untruncated* arithmetic error, whose fourth moment is only controlled here in Theorem 7.1.

What remains to be supplied.

1. **A fully justified weighted explicit formula.** The formula relating $P_X^W - S$ to contributions of zeros, the pole and any remaining terms must be established with uniformity in T and in arbitrarily large X, including the treatment of jumps and of t near γ_k. The corresponding proposition in [3] is presently a sketch.

2. **An arithmetic fourth-moment estimate beyond $X \leq T^{1/4}$.** The bound (9.3) requires cancellation among prime powers beyond what the direct Montgomery–Vaughan argument supplies. A fourth-moment estimate on long intervals need not be replaced by a uniform mean square on windows of length one: the latter is sufficient in some approaches but is not logically necessary.

3. **Non-cancellation for off-critical zeros.** The abstract model in Theorem 4.1 cannot establish the behavior of the actual P_X^W. A genuine implication towards RH needs a proved saturation or alternative obstruction when $\beta_k > 1/2$, with no unjustified hypothesis that a finite set of zeros achieves an isolated maximal real part.

4. **Separate cell containment from convergence.** A sequence of strictly increasing $\tau_k^{\mathrm{Cms}}(X;W)$ can converge as $X \to \infty$ to a nondecreasing sequence with equal consecutive values. Thus (5.6) alone does not directly establish RH or simplicity by an argument based solely on strict increase at finite X.

10 Exact sublevel identity and the stronger cell condition

The following identity is valid for every measurable F with finite sublevel measures, irrespective of RH:

$$u_k(F) - \gamma_k = \int_0^\infty (\mathbf{1}_{\{F(t)<k-\frac{1}{2}\}} - \mathbf{1}_{\{N(t)<k-\frac{1}{2}\}})\,\mathrm{d}t. \tag{10.1}$$

Indeed $\{N < k - \frac{1}{2}\}$ equals $(0, \gamma_k)$ up to a set of measure zero. For $F = F_X^W$, write the right-hand side as

$$\mathcal{A}_k^+(X) = \mathrm{meas}\{t : F_X^W(t) < k - \tfrac{1}{2} \le N(t)\}, \tag{10.2}$$

$$\mathcal{A}_k^-(X) = \mathrm{meas}\{t : N(t) < k - \tfrac{1}{2} \le F_X^W(t)\}, \tag{10.3}$$

so that $\tau_k^{\mathrm{Cms}}(X;W) - \gamma_k = \mathcal{A}_k^+(X) - \mathcal{A}_k^-(X)$. A global L^4 bound concerns the *size* of the arithmetic discrepancy, not uniform cancellation between the two signed sublevel contributions for every k.

For any strictly increasing sequence u_k, put $\bar{u}_k = (u_k + u_{k+1})/2$.

Proposition 10.1 (A sufficient cell condition for RH and simplicity)**.** *If, with a suitable first cell,*

$$\frac{u_{k-1} + u_k}{2} < \gamma_k < \frac{u_k + u_{k+1}}{2} \qquad (k \ge 2), \tag{10.4}$$

then every nontrivial zero has real part $\frac{1}{2}$ and is simple.

Proof. The intervals in (10.4) are disjoint. Consequently consecutive γ_k cannot be equal. A zero of multiplicity greater than one would repeat its imaginary part. A zero $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ would have the distinct reflected zero $1 - \beta + i\gamma$, again repeating the imaginary part. Both cases are excluded. □

Remark 10.2. The weaker condition $\gamma_{k-1} \le u_k < \gamma_{k+1}$ does not imply RH or simplicity; it can accommodate equal consecutive γ_k. The proposition is a sufficient criterion, *not* a claim that the required cell containment has been proved for the proposed taucm(s).

11 Finite computations recorded in the associated research files

For context, the author's research notes [16] for the branch `kc-tauC-2026-11-02` of the repository `vcasana1/prueba` record the following root-mean-square discrepancies, normalized by the mean spacing between successive γ_k, over the first 2,001,052 zeros. These are *reported measurements*, not computations independently repeated or certified for this note; they do not imply a uniform-in-k result.

Approximation scheme	Reported RMS
Main term N_0 alone	0.332
Original IJNT τ_k^{C}	0.0557
Sharp prime cutoff at $p \le t^{1.4}$	0.01456
Selberg weight with $X = t^{0.7}$ (support to $X^2 = t^{1.4}$)	0.01456
$W_{1/4}$ with $X = t^{0.7}$ (same support)	**0.01376**

The notes also report that the $W_{1/4}$ weight reduces RMS by approximately 5.5% relative to the sharp cutoff at the same support, but performs less well for the two exceptionally close pairs indexed 4765–4766 and 6709–6710. These measurements concern earlier prime-phase approximations; no new numerical estimate for τ_k^{Cm} under the MCOM truncation bound is claimed. Importantly, $X = t^{0.7}$ lies *outside* the rigorously uniform range $X \le T^{1/4}$ proved in Theorem 7.1. This distinction prevents a numerical experiment from being mistaken for an extension of that theorem.

12 Separation from the Riemann–von Mangoldt points

Let γ_k^{RvM} be the unique solutions of $N_0(\gamma_k^{\mathrm{RvM}}) = k - \frac{1}{2}$ with $\gamma_k^{\mathrm{RvM}} > 2\pi$. Since $N_0'(t) = (2\pi)^{-1}\log(t/2\pi)$,

$$\gamma_{k+1}^{\mathrm{RvM}} - \gamma_k^{\mathrm{RvM}} = \frac{1}{N_0'(\upsilon_k)} > 0 \tag{12.1}$$

for some υ_k between the two points, and this difference is $\sim 2\pi/\log\gamma_k^{\mathrm{RvM}}$ as $k \to \infty$. For the weighted constructions, put

$$\Delta_k(X;W) = \tau_{k+1}^{\mathrm{Cms}}(X;W) - \tau_k^{\mathrm{Cms}}(X;W) = \mathrm{meas}\{t > 0 : k - \tfrac{1}{2} \le F_X^W(t) < k + \tfrac{1}{2}\}. \tag{12.2}$$

The identity is immediate from the definition of sublevel measure. A continuous function attaining both levels gives $\Delta_k(X;W) > 0$; for sufficiently large k this follows from $F_X^W = N_0 + O(H)$ and $N_0 - H \to \infty$, uniformly in X. For a small index, the existence of a point below the lower level must be checked: continuity and the upper bound alone do not guarantee strict positivity for every X. Even if all finite-cutoff spacings are positive, $\lim_{X\to\infty}\Delta_k(X;W)$ may be zero.

Theorem 12.1 (Uniform separation of sufficiently distant indices)**.** *For k sufficiently large, let a_k, b_k be the unique solutions in the eventual increasing region of $N_0(t) + H(t) = k - \frac{1}{2}$ and $N_0(t) - H(t) = k - \frac{1}{2}$, respectively. Then*

$$a_k \le \tau_k^{\mathrm{Cms}}(X;W) \le b_k \qquad (X \ge 2). \tag{12.3}$$

For every positive integer m satisfying $m > 2H(b_k)$,

$$\inf_{X\ge 2}\left[\tau_{k+m}^{\mathrm{Cms}}(X;W) - \tau_k^{\mathrm{Cms}}(X;W)\right] \ge a_{k+m} - b_k > 0. \tag{12.4}$$

The same conclusion applies to taucm.

Proof. The inequalities $N_0 - H \le F_X^W \le N_0 + H$ and the eventual monotonicity of both barriers show that the sublevel measure is at least a_k and at most b_k, once the finitely many small-t values cannot reach the levels in question. Further, $N_0(b_k)+H(b_k) = k-\frac{1}{2}+2H(b_k) < k+m-\frac{1}{2}$. Strict increase of $N_0 + H$ implies $b_k < a_{k+m}$; subtract (12.3) at the two indices. □

Remark 12.2 (Asymptotic MCOM improvement). For large k the sufficient index separation is $m > (0.16764 + o(1))\log\gamma_k^{\mathrm{RvM}} + O(1)$. Using a non-explicit large-height truncation bound obtained from the MCOM asymptotic coefficient, the coefficient can be replaced by $2C_{\mathrm{H}} + o(1) = 0.1497192285422\ldots + o(1)$, where $C_{\mathrm{H}} = 0.0748596142710907\ldots$. Neither statement covers adjacent indices $m = 1$ for all large k.

Proposition 12.3 (Two local arithmetic occupation tests)**.** *For k sufficiently large put*

$$\begin{aligned} I_k^+ &= \{t > 0 : k - \tfrac{1}{2} < N_0(t) + H(t) < k + \tfrac{1}{2}\},\\ I_k^- &= \{t > 0 : k - \tfrac{1}{2} < N_0(t) - H(t) < k + \tfrac{1}{2}\},\\ B_k^+(X;W) &= \mathrm{meas}\{t \in I_k^+ : P_X^W(t) \ge H(t)\},\\ B_k^-(X;W) &= \mathrm{meas}\{t \in I_k^- : P_X^W(t) \le -H(t)\}. \end{aligned}$$

Then $I_k^+ \cap I_k^- = \varnothing$ *for large* k *and*

$$\boxed{\Delta_k(X;W) \geq B_k^+(X;W) + B_k^-(X;W).} \tag{12.5}$$

Consequently, $\limsup_{X\to\infty}(B_k^+ + B_k^-) > 0$ *is a sufficient arithmetic condition for noncollapse at index* k*.*

Proof. On I_k^+, the condition $P_X^W \geq H$ makes $F_X^W = N_0 + H \in (k - \frac{1}{2}, k + \frac{1}{2})$; the negative case is analogous. Both sets therefore contribute to (12.2). Since $2H > 1$ in the large-height region, the two intervals are disjoint. □

Remark 12.4 (Relevance and exact open issue). The occupation inequality uses only N_0, the MCOM truncation and Euler–Selberg phases. It does not establish that either B_k^+ or B_k^- stays positive as $X \to \infty$. The structural saturation model of Theorem 4.1 *does* have nonzero limiting spacings; a single increasingly sharp passage from $N_0 - H$ to $N_0 + H$ can, in contrast, make one spacing vanish. Therefore the separation of γ_k^{RvM} cannot alone be transferred to the τ_k^{Cms}. If DBSRH(W) and $\limsup_{X\to\infty} \Delta_k(X;W) > 0$ hold for every k, then $\gamma_{k+1} > \gamma_k$ for all k and the functional equation implies RH and simplicity. This implication is conditional on *both* hypotheses, not a proof of either.

13 The exact index defect at zero heights

To prevent a notation collision with the sublevel measure $u_k(F)$ from (1.4), denote the *evaluation defect* by

$$e_k(F) = k - \tfrac{1}{2} - F(\gamma_k). \tag{13.1}$$

For integer $j \geq 1$ define

$$\Psi_j(z) = \int_{z-1/2}^{z+1/2} v^{2j}\, dv = \frac{(z+1/2)^{2j+1} - (z-1/2)^{2j+1}}{2j+1}. \tag{13.2}$$

In particular,

$$\Psi_1(z) = z^2 + \frac{1}{12}. \tag{13.3}$$

The IJNT exact-defect identity relates an integral of $(N - F)^{2j}F'$ to the sum of $\Psi_j(e_k(F))$, with explicit terms at the two integration endpoints. When applying that identity, the endpoint convention and the factor $F'(t)$ must not be omitted, especially if F is nonmonotone. The exact form, with its endpoint terms, is reproduced in (13.7) at the end of this section.

Proposition 13.1 (Repeated imaginary parts force an evaluation defect)**.** *Suppose* F *is a single-valued real function at all the* γ_k*. If* $\gamma_{k+1} = \gamma_k$*, then*

$$e_{k+1}(F) - e_k(F) = 1, \qquad e_k(F)^2 + e_{k+1}(F)^2 \geq \frac{1}{2}. \tag{13.4}$$

More generally, among any finite group of $m \geq 2$ *equal imaginary parts,* $\sum e_k(F)^2 \geq m(m^2 - 1)/12$*. In any consecutive index window, let* M *count the indices lying in fully contained repeated-height groups. Then*

$$M \leq 4 \sum_{k \text{ in window}} e_k(F)^2. \tag{13.5}$$

In particular, if $|e_k(F)| < \frac{1}{2}$ *for each index of the window, no two* consecutive indices inside the window *can have the same imaginary part.*

Proof. At a shared imaginary part, the value of F is unchanged, so $e_{k+1} = e_k + 1$. The minimum of $z^2 + (z+1)^2$ is $1/2$. For m successive values $z, z+1, \ldots, z+m-1$, the sum of squares is minimized at $z = -(m-1)/2$, giving $m(m^2-1)/12$. Since $m \le 4m(m^2-1)/12$ for $m \ge 2$, summing over distinct repeated-height groups proves (13.5). The strict cell statement follows immediately from $e_{k+1} - e_k = 1$. □

Remark 13.2 (Finite-window and high-moment interpretation). For a group of two coincident imaginary parts, $\Psi_j(e_k) + \Psi_j(e_{k+1}) \ge 2/(2j+1)$, whereas the two baseline terms contribute $2 \cdot 4^{-j}/(2j+1)$. Their excess is at least $2(1-4^{-j})/(2j+1)$. This argument rigorously applies to *disjoint* coincident pairs. Counting overlapping pairs with the same coefficient requires a separate combinatorial proof. For a finite window, strict cell inequalities $|e_k(F)| < 1/2$ imply absence of internal coincidences. They do not imply the inequalities for all zero heights, nor do high-moment estimates at a fixed prime budget settle the all-height problem.

Proposition 13.3 (Forward evaluation: the proved implication)**.** *Let $\mathscr{F}_X(t)$ be any family of real-valued functions, and suppose*

$$\lim_{X\to\infty} \mathscr{F}_X(\gamma_k) = k - \tfrac{1}{2} \qquad \text{for every } k \ge 1. \tag{13.6}$$

Then RH holds and every nontrivial zero is simple.

Proof. If two consecutive zero indices share the same imaginary part, $\mathscr{F}_X$ has the same value at both indices for every X, contradicting the two different required limits. Thus all imaginary parts are distinct, which excludes both multiple zeros and, by the functional equation, any zero with real part different from $1/2$. □

Remark 13.4. For the proposed family $\mathscr{F}_X = N_\vartheta + P_X^W$, with $N_\vartheta(t) = \vartheta(t)/\pi + 1$, the converse implication (RH and simplicity ⇒ (13.6)) requires a rigorous pointwise weighted explicit formula evaluated at the zeros. It is not proved here and must not be replaced by an unverified midpoint-kernel argument. The same caution applies to the suggested uncorrected evaluations at a hypothetical off-critical zero.

The exact defect identity with its endpoint terms. For indices $1 \le k_1 < k_2$ and a real absolutely continuous F on a compact interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, the IJNT manuscript [1] (Theorem "Exact defect identity", label `thm:identity`) gives, for every integer $j \ge 1$ and with $e_k = e_k(F)$,

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \big(N(t) - F(t)\big)^{2j} F'(t)\, dt = \sum_{k=k_1+1}^{k_2-1} \Psi_j(e_k) + \frac{(\frac{1}{2} + e_{k_1})^{2j+1} + (\frac{1}{2} - e_{k_2})^{2j+1}}{2j+1}. \tag{13.7}$$

No monotonicity of F is required, and repeated imaginary parts are admitted: a degenerate gap $\gamma_k = \gamma_{k+1}$ contributes zero to both sides.

14 Numerical evidence from the 2,001,052-zero dataset

The following results are *reported numerical computations*, corroborated here against the text outputs in the repository directory `KC_tauC_2026-11-02/v8_identidad_en_ordenadas`. The binary arrays were not independently recomputed and the NUFFT/`float64` calculations have not been certified with interval arithmetic. In particular "passes the cell test numerically" does not mean a new rigorous verification of RH. The computation uses the evaluation defects (13.1) for three specific prime-phase rules, with $X = t^\theta$ for the weighted rules. The numerical variable θ is a prime-support exponent; it is not the Riemann–Siegel function ϑ.

Rule	RMS e_k	η on $(2^{19}, 2^{20}]$	$1-4\eta$	$\lvert e_k\rvert \geq 1/2$
IJNT original	0.0910	0.00840	0.9664	220
$W_{1/4}$, $\theta = 0.5$	0.0569	0.00330	0.9868	3
$W_{1/4}$, $\theta = 0.7$	**0.0381**	0.00149	**0.9941**	0

Here η is the average of $e_k(F)^2$ in the indicated dyadic height block; $1-4\eta$ is the numerical value of the lower bound that Proposition 13.1 would supply for the fraction of single-index-height zeros if the e_k values were certified. These values are *not* unconditional asymptotic proportions.

Rule	Passing / 2001	median/max j^*	minimum margin
IJNT original	1790 (89.46%)	7/23	negative (211)
$W_{1/4}$, $\theta = 0.5$	1998 (99.85%)	5/10	negative (3)
$W_{1/4}$, $\theta = 0.7$	2001 (100%)	4/7	0.0117

Windows comprise 1000 consecutive indices each. The numerical script checks $\max|e_k| < 1/2$ before computing a sufficient high-moment threshold j^* for each window. The final 52 indices of the dataset are not part of these 2001 complete windows; in the $\theta = 0.7$ run the maximum $|e_k|$ across the *full* dataset remains below $1/2$. The label "not passing" does not assert that the high-moment inequality fails for every j, only that this sufficient implementation rejects the window when an evaluation lies outside the strict cell.

Let g_k denote the nearest-neighbor separation normalized by the local mean separation $2\pi/\log(\gamma_k/2\pi)$, and put $\delta_k = 1/2 - |e_k(F)|$. For the 510 reported points with $g_k < 0.05$, least-squares fits are

$$\delta_k \approx \begin{cases} 1.49\, g_k - 0.0002 & \text{original IJNT}, \\ 1.79\, g_k - 0.0003 & W_{1/4},\ \theta = 0.5, \\ 2.50\, g_k + 0.0001 & W_{1/4},\ \theta = 0.7. \end{cases} \tag{14.1}$$

The slopes exhibit the numerical ratio $2.50/1.79 \approx 1.40$, matching $0.7/0.5 = 1.4$ to the reported precision; this is evidence for, not a proof of, linear dependence on θ. The average $\int_1^2 v(v/2 + 1/4)\, dv = 37/24 \approx 1.542$ is consistent with the proposed leading dependence $2\theta \int vq(v)\, dv$, but the $O(1)$ terms and other zeros have not been rigorously controlled. For the closest reported pair ($k = 1{,}115{,}578, 1{,}115{,}579$; $g_k \approx 0.005$), the two defects for $\theta = 0.7$ are -0.4883 and $+0.4846$. The 1% lower-fit intercepts are approximately $-0.1395, -0.0642, -0.0246$ for the three rules.

The extrapolation $\theta_{\text{needed}}(n) \approx 0.21 n^{0.09}$ uses a GUE-inspired minimum-gap law together with two measured choices of θ; it is a heuristic only, not a theorem, and it cannot justify applying mean-value estimates beyond their proved length ranges. In particular, the weighted cutoff $X = t^{0.5}$ has prime-power support $n \leq t$, well beyond the $X \leq T^{1/4}$ condition proved for Theorem 7.1, and even the logarithmically enlarged range of Theorem 7.3. The original IJNT rule retains a different prime budget: approximately $\sqrt{t}$ *primes*, not weighted prime powers supported up to t.

15 The $j = 1$ conditional proportion budget: new finite computations

The updated repository outputs [15] record the exact-defect identity at $j = 1$ in a form which distinguishes two contributions that must *not* be conflated. Here $F = N_0 + \Phi_c$, where Φ_c is the

specific truncated weighted phase used in the numerical script and $D_F(t) = N(t) - F(t)$. Write

$$\begin{aligned} M_T &= N(2T) - N(T), \\ A_T(F) &= \int_T^{2T} D_F(t)^2 N_0'(t)\, dt, \\ B_T(F) &= \int_T^{2T} D_F(t)^2 \Phi_c'(t)\, dt. \end{aligned} \tag{15.1}$$

The derivative Φ_c' is the almost-everywhere derivative, including the derivative of the moving truncation bound on its active portions. In particular, *it is not permissible to omit* B_T. Define $\eta_T(F) = M_T^{-1} \sum_{T<\gamma_k\le 2T} e_k(F)^2$. The exact IJNT identity gives, with its precise endpoint contribution $\mathcal{E}_\partial(F;T)$,

$$\eta_T(F) = \frac{A_T(F) + B_T(F)}{M_T} - \frac{1}{12} + \mathcal{E}_\partial(F;T). \tag{15.2}$$

Here the sign of $\mathcal{E}_\partial$ is *defined* by (15.2); the boundary terms in (13.7) are required if one claims a numerical upper bound for it. The numerical comparison between the two sides is a consistency test, not an independent analytic control of the endpoint contribution.

Dyadic T	A_T/M_T	B_T/M_T	$A_T/M_T + B_T/M_T - 1/12$	measured η_T
2^{10}	0.04647	+0.03883	0.00196	0.00200
2^{15}	0.05064	+0.03546	0.00277	0.00281
2^{19}	0.05253	+0.03407	0.00327	0.00330

The table is copied from `decomp_w14_th0.5.txt`. The phase in these computations uses $W_{1/4}$ at support exponent $\theta = 1/2$ and the original $\log\log t$ truncation bound: these numbers are *not* measurements for $H(t)$ of MCOM, nor does this table establish a bound as $T \to \infty$. The displayed discrepancies are at most $5 \cdot 10^{-5}$ at the reported precision.

Theorem 15.1 (Conditional lower bound for the simple critical-line fraction)**.** *Let $F = N_0 + \Phi_c$ be locally absolutely continuous, so that the exact identity (15.2) holds on dyadic intervals. For T sufficiently large, suppose that*

$$\begin{aligned} &T^{-1}\int_T^{2T} |D_F(t)|^2\, dt \le a_T, \quad T^{-1}\int_T^{2T} |D_F(t)|^4\, dt \le b_T, \\ &\frac{\left(T^{-1}\int_T^{2T} |\Phi_c'(t)|^2 dt\right)^{1/2}}{T^{-1}\int_T^{2T} N_0'(t) dt} \le \kappa_T, \qquad |\mathcal{E}_\partial(F;T)| \le r_T. \end{aligned} \tag{15.3}$$

Then the fraction of indices k with $T < \gamma_k \le 2T$ whose zeros are simple and on $\Re s = 1/2$ is at least

$$\boxed{1 - 4\max\Big(0,\ (1 + O(1/\log T))\,[a_T + \kappa_T\sqrt{b_T}] - \tfrac{1}{12} + r_T\Big) - O\left(\frac{\log T}{M_T}\right)} \tag{15.4}$$

The constants implicit in $O(1/\log T)$ are absolute. This is a conditional estimate on the three explicitly specified moments*, not an unconditional improvement of any published percentage.*

Proof. By (13.5), at most $4\sum e_k(F)^2$ indices in fully contained groups of equal imaginary parts can be non-simple or associated with an off-critical zero. The functional equation pairs any off-critical zero with a distinct zero at the same imaginary part; thus every excluded index belongs to such a group. Only the at most two groups intersecting the endpoints of the

window need separate treatment. Their total multiplicities are $O(\log T)$, using the unconditional $S(t) = O(\log t)$ and the zero-counting formula. This gives a simple-critical-line fraction at least $1 - 4\eta_T(F) - O(\log T/M_T)$. Moreover

$$\frac{A_T(F)}{M_T} \le \frac{T \sup_{[T,2T]} N_0'}{M_T} a_T = (1 + O(1/\log T))a_T.$$

By Cauchy–Schwarz,

$$\frac{|B_T(F)|}{M_T} \le \frac{T}{M_T}\sqrt{b_T}\left(\frac{1}{T}\int_T^{2T} |\Phi_c'(t)|^2 dt\right)^{1/2} = (1 + O(1/T))\kappa_T\sqrt{b_T}.$$

Insert these estimates and the endpoint bound into (15.2). Since $\eta_T(F) \ge 0$, the resulting upper bound for $\eta_T(F)$ may be replaced by its positive part. □

Remark 15.2 (Numerical budget and its logical status). For the measured block $T = 2^{19}$, the outputs `budget_w14_th0.5_j19.txt` and `mv_dphi_th0.5_j19.txt` report

$$a_{\mathrm{obs}} = 0.05253, \qquad b_{\mathrm{obs}} = 0.00813, \qquad \kappa_{\mathrm{model}} = 0.8241.$$

Substitution into the leading expression in (15.4) gives approximately 82.6%. This value is a *numerical conditional budget*; it is neither a new verified lower bound at infinite height nor a rigorous finite-height certificate. In particular, the cited Montgomery–Vaughan–Preissmann calculation treats a fixed-coefficient approximation to Φ_c'. The actual dyadic interpolation, the t-dependence of the weights, the pole compensator (if included), and derivatives of the active truncation introduce additional terms that must be bounded before the model constant 0.8241 may be used as a verified bound for κ_T. If in addition one could establish $b_T \le 3a_T^2$, $r_T = o(1)$, and $\kappa_T \le 0.8241 + o(1)$ uniformly as $T \to \infty$, then exceeding a proportion 0.417 would follow from $a_T < 0.09437\ldots$ eventually, while a positive proportion would follow from $a_T < 0.13732\ldots$. These thresholds are *algebraic consequences of the conditional assumptions*. The observed $a_{\mathrm{obs}} = 0.05253$ is not a proof of either bound.

Next analytic target. For the actual $W_{1/4}$ family with $\theta = 1/2$, derive explicit, height-uniform estimates for a_T, b_T, κ_T and the endpoint term r_T which satisfy (15.4). The estimate of the fourth moment with an unspecified implied constant is insufficient. The new calculation suggests that B_T is about 40% of the sum $A_T + B_T$ on the measured high block, so dropping it would be a substantial mathematical error. This approach seeks a new result about the author's *prime-corrected construction* via its exact defect identity, not a stand-alone classical characterization of RH.

16 Neighboring pairs: exact margins and finite-cutoff separation

The difference between a strict spacing at each finite cutoff and a nonzero limiting spacing is essential. This section studies that question for the arithmetic sublevel points themselves. The symbol $e_k(F)$ in (13.1) remains the *evaluation defect*; here the position error is denoted by $d_k(X; W)$ to avoid a collision. For k in the eventually increasing region of N_0 set

$$d_k(X; W) := \tau_k^{\mathrm{Cms}}(X; W) - \gamma_k, \qquad g_k := \gamma_{k+1} - \gamma_k \ge 0. \tag{16.1}$$

Proposition 16.1 (Exact symmetric-margin identity). *Define the symmetric margin*

$$\mathcal{M}_k(X; W) := \Delta_k(X; W) - |d_k(X; W)| - |d_{k+1}(X; W)|. \tag{16.2}$$

Writing $x_+ := \max(x, 0)$*, one has exactly*

$$\boxed{\mathcal{M}_k(X;W) = g_k - 2\big[(d_k(X;W))_+ + (-d_{k+1}(X;W))_+\big] \le g_k.} \tag{16.3}$$

Equality holds precisely when the two sublevel points enclose the corresponding imaginary parts, i.e. $\tau_k^{\mathrm{Cms}}(X;W) \le \gamma_k$ *and* $\tau_{k+1}^{\mathrm{Cms}}(X;W) \ge \gamma_{k+1}$*. Consequently* $\mathcal{M}_k(X;W) > 0$ *implies* $g_k > 0$*. Here* g_k *is the unnormalized gap of* (16.1)*, not the normalized nearest-neighbour separation of Section 14.*

Proof. By definition $\Delta_k = g_k + d_{k+1} - d_k$. Use $x - |x| = -2(-x)_+$ and $-x - |x| = -2x_+$. The criterion for equality follows because both terms subtracted from g_k are nonnegative. □

Remark 16.2 (What a numerical margin does not certify). The sublevel points are prime-defined, whereas (16.2) uses the true γ_k. Thus positivity of a margin computed from verified zeros is a useful finite test, not by itself a prime-only inequality valid at every height. A repeated imaginary part gives $g_k = 0$ and hence $\mathcal{M}_k \le 0$, even if $\Delta_k(X;W) > 0$ at every finite X.

Proposition 16.3 (Quantitative separation at a fixed prime cutoff)**.** *Fix* $X > 1$ *and* k *sufficiently large for the two barriers in Theorem 12.1 to be defined. Set*

$$\mathscr{L}_{k,X} := \operatorname*{ess\,sup}_{t\in[a_k,b_{k+1}]} |(F_X^W)'(t)|. \tag{16.4}$$

Then $0 < \mathscr{L}_{k,X} < \infty$ *and*

$$\boxed{\Delta_k(X;W) \ge \mathscr{L}_{k,X}^{-1} > 0.} \tag{16.5}$$

This bound is finite-cutoff only: the hypotheses do not give an upper bound on $\mathscr{L}_{k,X}$ *independent of* X*.*

Proof. By the barriers, F_X^W attains a value at or below $k - \frac{1}{2}$ in the initial part of $[a_k, b_{k+1}]$ and a value at or above $k + \frac{1}{2}$ in its final part. Let r be a point of the initial part with $F_X^W(r) \le k - \frac{1}{2}$, put $s_1 = \inf\{t > r : F_X^W(t) \ge k + \frac{1}{2}\}$ and $s_0 = \sup\{t \in [r, s_1] : F_X^W(t) \le k - \frac{1}{2}\}$. By continuity and these extremal choices, $F_X^W(s_0) = k - \frac{1}{2}$, $F_X^W(s_1) = k + \frac{1}{2}$, and $k - \frac{1}{2} < F_X^W < k + \frac{1}{2}$ on (s_0, s_1). The Lipschitz bound gives $1 \le \mathscr{L}_{k,X}(s_1 - s_0) \le \mathscr{L}_{k,X}\Delta_k$. In particular $\mathscr{L}_{k,X} > 0$: a locally absolutely continuous function whose derivative vanishes almost everywhere on $[s_0, s_1]$ would be constant there. On a compact interval the prime phase, compensator and H are locally Lipschitz for fixed X; so is the truncation, hence $\mathscr{L}_{k,X} < \infty$. □

17 The $W_{1/4}$ two-source kernel: an exact resolution law

The following kernel analysis is a *local two-source model*; it is not an unconditional decomposition of the actual Euler phase into individual contributions of zeta zeros. Define

$$q_{1/4}(v) = \frac{v}{2} + \frac{1}{4} \quad (1 \le v \le 2), \qquad K(z) = \frac{1}{2} + \frac{1}{\pi}\int_1^2 q_{1/4}(v)\,\mathrm{Si}(vz)\,dv, \tag{17.1}$$

where $\mathrm{Si}(x) = \int_0^x (\sin y)/y\,dy$ is continuous at $x = 0$. The index $1/4$ labels the weight, not the real part of any zero. For $m \ge 0$, write

$$\mu_m := \int_1^2 v^m q_{1/4}(v)\,dv. \tag{17.2}$$

In particular $\mu_1 = 37/24$, $\mu_2 = 59/24$, $\mu_3 = 323/80$ and $\mu_4 = 34/5$.

Lemma 17.1 (Bounded derivatives of the two-source kernel)**.** *The function K extends to an entire function of z. On the real axis,*

$$\begin{aligned} K'(z) &= \frac{1}{\pi}\int_1^2 q_{1/4}(v)\int_0^v \cos(zy)\,dy\,dv, \\ K''(z) &= -\frac{1}{\pi}\int_1^2 q_{1/4}(v)\int_0^v y\sin(zy)\,dy\,dv, \\ K^{(4)}(z) &= \frac{1}{\pi}\int_1^2 q_{1/4}(v)\int_0^v y^3\sin(zy)\,dy\,dv. \end{aligned} \tag{17.3}$$

Thus $K(-z) = 1 - K(z)$ and

$$\sup_{z\in\mathbb{R}}|K''(z)| \le \frac{\mu_2}{2\pi}, \qquad \sup_{z\in\mathbb{R}}|K^{(4)}(z)| \le \frac{\mu_4}{4\pi}. \tag{17.4}$$

Moreover $K'(z), K''(z) \to 0$ as $|z| \to \infty$ through real values, and neither is identically zero.

Proof. Since $\mathrm{Si}(x) = \sum_{n\ge 0}(-1)^n x^{2n+1}/((2n+1)(2n+1)!)$ is entire and the v-integral is over a compact set, K is entire. The identity $\mathrm{Si}(vz) = \int_0^v(\sin(zy))/y\,dy$, whose integrand has a removable singularity at $y = 0$, permits termwise differentiation on compact subsets of the complex plane. All estimates follow by integrating y and y^3 against the nonnegative weight. The decay of K' and K'' is the Riemann–Lebesgue lemma for the integrable, compactly supported kernels obtained after interchanging the v, y integrals, namely $K'(z) = \frac{1}{\pi}\int_0^2 w(y)\cos(zy)\,dy$ and $K''(z) = -\frac{1}{\pi}\int_0^2 yw(y)\sin(zy)\,dy$, with w as in the proof of Theorem 17.3. Finally $K'(0) = \mu_1/\pi > 0$, while $K'''(0) = -\mu_3/(3\pi) < 0$. □

Theorem 17.2 (Global quadratic resolution estimate)**.** *For real z and h, put $C(z,h) := K(z - h/2) + K(z + h/2)$. Then*

$$\boxed{C(z,h) = 2K(z) + \frac{h^2}{4}K''(z) + \mathcal{R}_K(z,h), \qquad |\mathcal{R}_K(z,h)| \le \frac{\mu_4}{768\pi}|h|^4 = \frac{17}{1920\pi}|h|^4.} \tag{17.5}$$

In addition,

$$\sup_{z\in\mathbb{R}}|C(z,h) - C(z,0)| \le \frac{\mu_2}{8\pi}h^2. \tag{17.6}$$

Consequently, with $h = Lg$ and $L = \log X$, a gap g small compared with $1/L$ changes the full two-source profile only at quadratic order L^2g^2. This is a theorem about *C, not an estimate of the contribution of all zeta zeros.*

Proof. Apply Taylor's theorem with fourth-order remainder to $K(z + h/2) + K(z - h/2)$. Odd powers cancel. The remainder is at most $2(|h|/2)^4 \sup|K^{(4)}|/4!$, proving (17.5); the second symmetric-difference inequality and (17.4) prove (17.6). □

Theorem 17.3 (Injectivity of the complete local pair profile)**.** *Fix $L > 0$ and define, for any center $c \in \mathbb{R}$ and separation $g \ge 0$,*

$$\mathcal{C}_{c,g,L}(t) := K(L(t - c - g/2)) + K(L(t - c + g/2))\,. \tag{17.7}$$

If $\mathcal{C}_{c_1,g_1,L}$ and $\mathcal{C}_{c_2,g_2,L}$ coincide on any nonempty open real interval, then $c_1 = c_2$ and $g_1 = g_2$. Thus, in the exact two-source model, the complete profile *identifies even an arbitrarily small positive gap. This is an injectivity statement, not a stable recovery estimate in the presence of unknown contributions.*

Proof. Define the compactly supported nonnegative function

$$w(y) = \int_{\max(1,y)}^{2} q_{1/4}(v)\,dv \quad (0 \le y \le 2).$$

For $0 < y < 1$, $w(y) = 1$. Interchanging the integrals in (17.3) gives

$$K'(z) = \frac{1}{\pi}\int_0^2 w(y)\cos(zy)\,dy.$$

Consequently,

$$\mathcal{C}'_{c,g,L}(t) = \frac{2L}{\pi}\int_0^2 w(y)\cos(L(t-c)y)\cos(Lgy/2)\,dy. \tag{17.8}$$

The profiles are entire in t, so agreement on an open real interval implies agreement everywhere, and then their derivatives agree. Writing $\cos(L(t-c)y)$ as the average of $e^{\pm iL(t-c)y}$ and substituting $y \mapsto -y$ in the second term,

$$\mathcal{C}'_{c,g,L}(t) = \frac{L}{\pi}\int_{-2}^{2} f_{c,g}(y)e^{iLty}\,dy, \qquad f_{c,g}(y) = w(|y|)\cos(Lg|y|/2)e^{-iLcy},$$

with $f_{c,g}$ integrable and supported in $[-2,2]$. Uniqueness of the Fourier transform of L^1 functions gives $f_{c_1,g_1} = f_{c_2,g_2}$ almost everywhere; both are continuous on $0 < y < 1$, where $w = 1$, so

$$e^{-iLc_1y}\cos(Lg_1y/2) = e^{-iLc_2y}\cos(Lg_2y/2), \qquad 0 < y < 1.$$

By analyticity this also holds at $y = 0$. Differentiate once at $y = 0$ to get $c_1 = c_2$, and twice to get $g_1^2 = g_2^2$. Since both separations are nonnegative, $g_1 = g_2$. □

Remark 17.4 (Why $\theta = 0.7$ is effective, without claiming optimality). If $X = t^\theta$ for the numerical weighted construction, the support of the prime-power phase extends to $n \le t^{2\theta}$. The choice $\theta = 0.7$ therefore reaches $t^{1.4}$, whereas $\theta = 0.5$ reaches t. This increases the local scale L and hence the scale $L^2g^2/4$ of the quadratic term in (17.5); the term itself still vanishes at the zeros of K''. The weight has $q_{1/4}(2) = 5/4$ and $\mu_1 = 37/24$; its upper endpoint receives more emphasis than in Selberg's constant density $q \equiv 1$. The observed superiority of $\theta = 0.7$ is empirical; neither its optimality nor an unconditional arithmetic separation theorem at that cutoff follows from this kernel calculation.

18 A prime-only spatial filter and its cutoff derivative

For the rest of this section, $X = e^L$ with $L > 0$ is a *fixed* cutoff and the phase is *untruncated*: $F_L^\circ(t) := N_0(t) + P_{e^L}^W(t)$. This differs from $F_{e^L}^W(t)$ in (5.5); replacing it by the clipped phase in a derivative identity would be an error. Here "prime-only" means built from prime powers through Λ, the pole compensator and N_0, without any information on the zeros. For a function G defined near $c > 0$ and $r > 0$ with $c > r/L$ set

$$\mathscr{D}_{L;r}[G](c) := \frac{G(c + r/L) - G(c - r/L)}{2r}. \tag{18.1}$$

For $0 < r_1 < r_2$ set

$$\mathscr{Q}_{L;r_1,r_2}[G](c) := \mathscr{D}_{L;r_1}[G](c) - \mathscr{D}_{L;r_2}[G](c), \qquad \mathscr{B}_{r_1,r_2}(v) := \frac{\sin(r_1v)}{r_1} - \frac{\sin(r_2v)}{r_2}. \tag{18.2}$$

Proposition 18.1 (Exact Euler formula for the spatial filter)**.** *For any admissible weight W of (5.1), the spatial filter of F_L° satisfies*

$$\begin{aligned}\mathscr{Q}_{L;r_1,r_2}[F_L^\circ](c) &= \mathscr{Q}_{L;r_1,r_2}[N_0](c)\\ &\quad - \frac{1}{\pi}\sum_{n\le e^{2L}} W\Big(\frac{\log n}{L}\Big)\,\frac{\Lambda(n)}{\sqrt{n}\log n}\cos(c\log n)\mathscr{B}_{r_1,r_2}\Big(\frac{\log n}{L}\Big)\\ &\quad - \mathscr{Q}_{L;r_1,r_2}[\Pi_{e^L}^W](c).\end{aligned} \tag{18.3}$$

Thus the right side is determined by the primes and the Riemann–von Mangoldt main term; there is no appearance of β_k or of the actual γ_k. If a background B is five times continuously differentiable on $[c-r_2/L, c+r_2/L]$, then

$$\begin{aligned}\mathscr{Q}_{L;r_1,r_2}[B](c) &= \frac{r_1^2-r_2^2}{6L^3}B'''(c)+\mathcal{R}_B,\\ |\mathcal{R}_B| &\le \frac{r_1^4+r_2^4}{120L^5}\sup_{|t-c|\le r_2/L}|B^{(5)}(t)|.\end{aligned} \tag{18.4}$$

In particular the constant and affine parts of B are eliminated exactly*, not merely approximately.*

Proof. The finite Euler sum has imaginary part

$$-\frac{1}{\pi}\sum_n \frac{\Lambda(n)}{\sqrt{n}\log n}W\Big(\frac{\log n}{L}\Big)\sin(t\log n).$$

Use the sine subtraction identity at $c\pm r/L$, then subtract the two radii. The compensator contributes the last term in (18.3). Taylor's theorem for the symmetric first difference gives $\mathscr{D}_{L;r}[B](c) = B'(c)/L + r^2B'''(c)/(6L^3) + \mathcal{R}$ with $|\mathcal{R}| \le r^4 \sup|B^{(5)}|/(120L^5)$. □

Proposition 18.2 (Cutoff derivative and the moving-point correction)**.** *Suppose q is continuous on $[1,2]$, with W given by (5.1). For $L>0$ away from the discrete values $L=\log n$ and $L=\frac12\log n$ (that is, $e^L=n$ or $e^{2L}=n$) of prime powers $n\ge 2$, write $\dot P_L(t) := (\partial/\partial L)P_{e^L}^W(t)$ with t* held fixed*. Then*

$$\boxed{L^2\dot P_L(t) = \frac{1}{\pi}\operatorname{Im}\left[\sum_{e^L<n<e^{2L}} q\Big(\frac{\log n}{L}\Big)\frac{\Lambda(n)}{n^{1/2+it}} - \int_{e^L}^{e^{2L}} q\Big(\frac{\log u}{L}\Big)\,u^{-1/2-it}\,du\right].} \tag{18.5}$$

The derivative is taken piecewise at the exceptional values and exists almost everywhere in L. For any family $G_L(t)$ differentiable in both variables, with G_L' the derivative in t and $\partial_L G_L$ the derivative in L, the total *derivative of the moving spatial filter is*

$$\begin{aligned}\frac{d}{dL}\mathscr{Q}_{L;r_1,r_2}[G_L](c) &= \mathscr{Q}_{L;r_1,r_2}[\partial_L G_L](c)\\ &\quad - \frac{1}{2L^2}\Big(G_L'(c+r_1/L)+G_L'(c-r_1/L)\\ &\qquad - G_L'(c+r_2/L)-G_L'(c-r_2/L)\Big).\end{aligned} \tag{18.6}$$

In particular, differentiating at fixed t and then taking the spatial filter is not *interchangeable with differentiating the filter as a function of L.*

Proof. Inside the transition region $1<\log n/L<2$, one has

$$\partial_L W\Big(\frac{\log n}{L}\Big) = \frac{\log n}{L^2}\,q\Big(\frac{\log n}{L}\Big);$$

outside this region the derivative is zero. For the pole compensator write $\Pi_{e^L}^W(t) = \frac{1}{\pi}\operatorname{Im}\int_{\log 2}^{\infty} W(v/L)e^{v/2-itv}\,\frac{dv}{v}$; for L in a compact neighbourhood of a non-exceptional value the integrand is supported in a fixed compact v-interval and its L-derivative is bounded there, so differentiation under the integral sign (dominated convergence) gives the integral term. This proves (18.5). For (18.6), differentiate $G_L(c\pm r/L)$ by the chain rule, retaining the moving-point contributions from $\mp r/L^2$, and subtract the two radii. □

19 Finite-sample identifiability and the remaining arithmetic error

The kernel calculation can be promoted to a precise *conditional* local inverse statement. It does not require GUE. It also does not supply, on its own, a decomposition of F_L° when RH is not assumed.

Theorem 19.1 (Nondegenerate quadratic local inverse)**.** *Fix any nonempty open interval $J \subset \mathbb{R}$. The six entire functions*

$$1, \quad z, \quad z^2, \quad z^3, \quad K'(z), \quad K''(z) \tag{19.1}$$

are linearly independent. Consequently there exist six distinct points $z_1, \dots, z_6 \in J$ such that the evaluation matrix of (19.1) is invertible. Let A be the 6×5 matrix obtained by evaluating the first five functions at these points, and let $\mathbf{v} = (K''(z_i))_{i=1}^6$. Let P be the orthogonal projection onto the orthogonal complement of the range of A, so that $\mathsf{P}\mathbf{v} \neq 0$. If measured prime-phase values, after subtraction of the coincident-pair model $2K$, admit the representation

$$\mathbf{y} = \mathsf{A}\mathbf{b} + \frac{h^2}{4}\mathbf{v} + \mathbf{r} + \mathbf{r}_4, \qquad \|\mathbf{r}_4\|_2 \le \frac{\mu_4}{768\pi}\sqrt{6}\,|h|^4, \tag{19.2}$$

where $\mathbf{b} \in \mathbb{R}^5$ is an unknown nuisance vector, $\mathbf{r}$ is an unspecified arithmetic residual, and $h = Lg \ge 0$, then the estimator, observable from the sampled vector $\mathbf{y}$ (and prime-observable once $\mathbf{y}$, the sample points, the centre and L are constructed from prime-defined data),

$$\widehat{h^2} := 4\frac{\langle \mathsf{P}\mathbf{y}, \mathsf{P}\mathbf{v}\rangle}{\|\mathsf{P}\mathbf{v}\|_2^2} \tag{19.3}$$

satisfies the rigorous conditional stability estimate

$$\boxed{|\widehat{h^2} - h^2| \le \frac{4}{\|\mathsf{P}\mathbf{v}\|_2}\left(\|\mathsf{P}\mathbf{r}\|_2 + \frac{\mu_4\sqrt{6}}{768\pi}|h|^4\right).} \tag{19.4}$$

A proved bound on the right smaller than a positive measured $\widehat{h^2}$ *would certify $h > 0$* provided *the representation (19.2) is valid for the actual phase.*

Proof. Suppose a linear combination of (19.1) vanishes on J. By analyticity it vanishes on all of $\mathbb{R}$. Its odd part gives $az + bz^3 + cK''(z) = 0$; since $K''(z) \to 0$ at infinity, $a = b = 0$, then $c = 0$ because $K'' \not\equiv 0$. Its even part gives $d + ez^2 + fK'(z) = 0$, hence $d = e = f = 0$ by decay and $K'(0) > 0$. Linear independence of analytic functions implies that evaluations at some six distinct points of any open interval span $\mathbb{R}^6$; otherwise the evaluation vectors would lie in a common proper subspace, giving a forbidden linear relation on J. Thus $\mathsf{P}\mathbf{v} \neq 0$. Apply P to (19.2), take the scalar product with $\mathsf{P}\mathbf{v}$ and use Cauchy–Schwarz. The bound on $\mathbf{r}_4$ follows from Theorem 17.2. □

Remark 19.2 (Center uncertainty and the limit of the result). A small shift of the assumed center changes $2K(z)$ by a term proportional to $K'(z)$ at first order, which explains the fifth column of A. Higher powers of the center error must be included in $\mathbf{r}$ or estimated separately. A smooth local background is not *automatically* a cubic polynomial, even when its derivatives exist; its interpolation residual is also included in $\mathbf{r}$. Most importantly, the off-critical-zero contribution to the full Euler phase cannot be declared part of a controlled background without proof. Thus (19.4) is an exact inverse stability theorem, *not* a proof that the actual prime residual satisfies the necessary bound. Explicitly, if a proved bound $\mathcal{B}$ for the right side of (19.4) satisfies $\mathcal{B} < \widehat{h^2}$, then $h^2 \geq \widehat{h^2} - \mathcal{B} > 0$. The estimator is an algebraic quantity and may be negative when a residual is present.

Research reduction for the actual weighted constructions. For every fixed k and a sequence of cutoffs $X \to \infty$, the new quantitative task is to establish a representation of the form (19.2) in a neighborhood selected using only the prime-defined sublevel points, together with a bound for $\|\mathsf{P}\mathbf{r}\|_2$ that is *smaller than the quadratic signal* for any positive true separation. Such a bound must be proved without imposing that the relevant zeros lie on the critical line; if this cannot be achieved, the local inverse gives an effective finite-height diagnostic but not RH. It is independent of whether any hypothetical off-line zero has a maximal real part.

20 Local spacing observations: scope

The following data are included solely to identify the regime motivating Sections 17 to 19. They are not numerically certified. Unfolded spacing here means applying N_0 to the corresponding positive heights, where N_0 is increasing. In the index range $10^6 \leq k < 2{\cdot}10^6$, the nearest-neighbor study reported:

Prime rule	Minimum gap	Median tauc gap for 122 tight pairs	Median midpoint error
IJNT original	0.235	0.354	0.0092
$W_{1/4}$, $\theta = 0.5$	0.222	0.293	0.0023
$W_{1/4}$, $\theta = 0.7$	0.173	0.210	0.0008
Sharp support, $\theta = 1.4$	0.144	0.163	0.0023

These numbers concern the numerical *original clipped phases* with the small $\log\log t$ truncation in the supplied scripts; they must not be silently reinterpreted as calculations for the MCOM-truncated F_X^W with the pole compensator. For gaps below 0.05 unfolded units, the reported Spearman correlation between the two tauc gaps and the real gaps was close to zero for all tested rules, whereas the midpoint error was much smaller than individual point errors. A separate margin study reports $\mathcal{M}_k = g_k$ for the tight pairs in the weighted $\theta = 0.5, 0.7$ constructions, as predicted by (16.3) when the two sublevel points enclose the actual pair. These observations motivate, but do not establish, the local quadratic model. In particular they do not imply that an off-line pair can be discriminated by a fixed prime cutoff.

21 Correlated displacement from the Riemann–von Mangoldt points: what proximity does and does not prove

This section gives the precise response to a natural question: since the Riemann–von Mangoldt points are strictly separated and the prime-defined sublevel points are strictly separated at each finite cutoff, can their *proximity* prevent two consecutive sublevel points from colliding as $X \to \infty$? The answer is a conditional yes, provided the *difference of consecutive displacements*

is controlled. Neither finite-cutoff strictness nor an amplitude bound alone proves the required control. The conclusions here concern the author's own constructions and do not introduce a separate classical criterion for RH.

21.1 Exact decomposition into a positive main spacing and a correlated arithmetic displacement

Let γ_k^{RvM} be the unique solution greater than 2π of $N_0(\gamma_k^{\mathrm{RvM}}) = k - \frac{1}{2}$, and put

$$h_k^{\mathrm{RvM}} := \gamma_{k+1}^{\mathrm{RvM}} - \gamma_k^{\mathrm{RvM}} > 0. \tag{21.1}$$

The positivity holds for every $k \geq 1$ since N_0 is strictly increasing on $(2\pi, \infty)$ and its range on that interval contains every $k - \frac{1}{2}$: indeed $N_0(2\pi) = -1 + \frac{7}{8} = -\frac{1}{8}$, so the range is $(-\frac{1}{8}, \infty)$. For the weighted, MCOM-truncated sublevel points (5.5), define the *RvM displacement*, distinct from the true-zero position error $d_k(X;W)$ in (16.1), by

$$\varepsilon_k^{\mathrm{RvM}}(X;W) := \tau_k^{\mathrm{Cms}}(X;W) - \gamma_k^{\mathrm{RvM}}. \tag{21.2}$$

It is determined by the primes, the pole compensator and N_0, without using any γ_k or β_k.

Proposition 21.1 (Exact correlated-displacement identity)**.** *For every $k \geq 1$ and every cutoff X for which the defining sublevel measures are finite,*

$$\boxed{\Delta_k(X;W) = h_k^{\mathrm{RvM}} + \varepsilon_{k+1}^{\mathrm{RvM}}(X;W) - \varepsilon_k^{\mathrm{RvM}}(X;W).} \tag{21.3}$$

In particular, for a fixed index k the following two properties are equivalent:

$$\liminf_{X\to\infty} \Delta_k(X;W) > 0, \tag{21.4}$$

$$\limsup_{X\to\infty} \left[\varepsilon_k^{\mathrm{RvM}}(X;W) - \varepsilon_{k+1}^{\mathrm{RvM}}(X;W)\right] < h_k^{\mathrm{RvM}}. \tag{21.5}$$

A sufficient quantified form is: for some $\eta_k \in (0,1)$ and $X_k > 1$,

$$\varepsilon_k^{\mathrm{RvM}}(X;W) - \varepsilon_{k+1}^{\mathrm{RvM}}(X;W) \leq (1-\eta_k) h_k^{\mathrm{RvM}} \qquad (X \geq X_k). \tag{21.6}$$

Under (21.6), $\Delta_k(X;W) \geq \eta_k h_k^{\mathrm{RvM}} > 0$ for $X \geq X_k$. There is no requirement that η_k be uniform in k.

Proof. Subtract (21.2) at the consecutive indices and use (21.1). Taking upper and lower limits yields

$$\liminf_{X\to\infty} \Delta_k(X;W) = h_k^{\mathrm{RvM}} - \limsup_{X\to\infty} (\varepsilon_k^{\mathrm{RvM}} - \varepsilon_{k+1}^{\mathrm{RvM}}),$$

with the usual extended-real interpretation if a limit is infinite. The quantified conclusion follows by direct substitution. □

Remark 21.2 (Exact identity, not an independent arithmetic bound). Condition (21.5) is an exact reformulation of noncollapse, not a proof of noncollapse. Its value is to identify *which arithmetic difference* one must estimate. Controlling $|\varepsilon_k^{\mathrm{RvM}}|$ and $|\varepsilon_{k+1}^{\mathrm{RvM}}|$ separately is sufficient if

$$|\varepsilon_k^{\mathrm{RvM}}| + |\varepsilon_{k+1}^{\mathrm{RvM}}| \leq (1-\eta_k) h_k^{\mathrm{RvM}},$$

but is stronger than necessary. A large common displacement cancels exactly in (21.3). The quantities $\varepsilon_k^{\mathrm{RvM}}$ must not be confused either with the evaluation defects $e_k(F)$ of (13.1) or with the true-zero errors $d_k(X;W)$ of (16.1).

21.2 What the existing MCOM truncation actually yields

The large-index barriers of Theorem 12.1 provide an *individual* proximity estimate. The asymptotic MCOM coefficient C_{H} must not be used for the *fixed* truncation H; the coefficient appropriate to H is computed here.

Proposition 21.3 (Uniform RvM proximity from the fixed truncation)**.** *Let*

$$H(t) = A_{\mathrm{M}} \log \max(t, e) + B_{\mathrm{M}}, \qquad A_{\mathrm{M}} = 0.08382, \quad B_{\mathrm{M}} = 5.5,$$

and let a_k, b_k be the large-index barriers of Theorem 12.1. Uniformly in $X \geq 2$,

$$-\frac{H(a_k)}{N_0'(\xi_{k,-})} \leq \varepsilon_k^{\mathrm{RvM}}(X; W) \leq \frac{H(b_k)}{N_0'(\xi_{k,+})}, \tag{21.7}$$

where $\xi_{k,-} \in (a_k, \gamma_k^{\mathrm{RvM}})$ and $\xi_{k,+} \in (\gamma_k^{\mathrm{RvM}}, b_k)$ are supplied by the mean-value theorem. Consequently,

$$\boxed{\sup_{X \geq 2} |\varepsilon_k^{\mathrm{RvM}}(X; W)| \leq 2\pi A_{\mathrm{M}} + o(1) \qquad (k \to \infty).} \tag{21.8}$$

On the other hand,

$$h_k^{\mathrm{RvM}} \sim \frac{2\pi}{\log \gamma_k^{\mathrm{RvM}}} \to 0. \tag{21.9}$$

Thus (21.8) does not imply either (21.5) or (21.4).

Proof. Set $v_k := \gamma_k^{\mathrm{RvM}}$. Since $N_0(a_k) + H(a_k) = N_0(v_k) = N_0(b_k) - H(b_k)$, the mean-value theorem gives

$$v_k - a_k = \frac{H(a_k)}{N_0'(\xi_{k,-})}, \qquad b_k - v_k = \frac{H(b_k)}{N_0'(\xi_{k,+})}.$$

The barriers $a_k \leq \tau_k^{\mathrm{Cms}}(X; W) \leq b_k$ yield (21.7). Moreover $a_k/v_k \to 1$ and $b_k/v_k \to 1$. For fixed $\delta \in (0, 1)$, $N_0(v_k) - N_0((1-\delta)v_k) \gg_\delta v_k \log v_k$, whereas $H((1-\delta)v_k) = O(\log v_k)$; hence $N_0 + H < k - \frac{1}{2}$ at $(1-\delta)v_k$, and $a_k > (1-\delta)v_k$ for large k because $N_0 + H$ is increasing there. Similarly $N_0((1+\delta)v_k) - N_0(v_k) \gg_\delta v_k \log v_k > H((1+\delta)v_k)$ gives $b_k < (1+\delta)v_k$. Since δ is arbitrary, the two ratios tend to 1. On this common asymptotic scale, $H(t) = A_{\mathrm{M}} \log v_k + O(1) + o(\log v_k)$ and $N_0'(t) = (\log v_k)/(2\pi) + O(1)$, so the two extremal differences are $2\pi A_{\mathrm{M}} + o(1)$. Finally, the mean-value theorem between v_k and v_{k+1} gives (21.9). □

Remark 21.4 (The asymptotic MCOM coefficient requires a different H). Suppose the author's asymptotic bound for $S(t)$ has coefficient $C_{\mathrm{H}} < A_{\mathrm{M}}$. For any fixed $c > C_{\mathrm{H}}$, one may *define a different sublevel family* whose truncation bound is eventually $\widetilde{H}_c(t) = c \log t$ and whose small-height extension is chosen sufficiently large to contain $|N(t) - N_0(t)|$. For that *different* family, the argument of Proposition 21.3 yields $\sup_X |\tilde{\varepsilon}_k^{\mathrm{RvM}}| \leq 2\pi c + o(1)$ as $k \to \infty$; one may then take $c \downarrow C_{\mathrm{H}}$ at the level of asymptotic coefficients. This does *not* replace A_{M} by C_{H} in (21.8) for the fixed $H(t) = 0.08382 \log \max(t, e) + 5.5$ of the present weighted taucms. Nor does the improved individual bound alone imply noncollapse.

21.3 What DBSRH would transfer, and what mean-square proximity cannot

Corollary 21.5 (Conditional transfer of RvM separation)**.** *Assume DBSRH(W) as formulated in (5.6) and, for every $k \geq 1$, condition (21.5). Then all positive imaginary parts of the zeros of ζ are pairwise distinct, so RH holds and every nontrivial zero is simple. Conversely,* under DBSRH(W)*, condition (21.5) for all k is equivalent to RH* together with *simplicity.*

Proof. By DBSRH(W), at each fixed index $\tau_k^{\mathrm{Cms}}(X;W) \to \gamma_k$. Therefore

$$\Delta_k(X;W) \longrightarrow \gamma_{k+1} - \gamma_k.$$

Proposition 21.1 makes the limit strictly positive for every k, hence no imaginary parts repeat. If a zero $\rho = \beta + i\gamma$ has $\beta \neq \frac{1}{2}$, the distinct zero $1 - \overline{\rho} = 1 - \beta + i\gamma$ repeats the same positive imaginary part. A multiple zero does so as well. Conjugation treats negative imaginary parts. This proves RH and simplicity. Conversely, if RH and simplicity hold, all consecutive γ_k are strictly increasing, and the displayed limit is positive. This implies (21.5) by Proposition 21.1. Thus, under DBSRH(W): (21.5) for every k $\iff$ $\gamma_{k+1} > \gamma_k$ for every k $\iff$ RH and simplicity. This is not an equivalence between DBSRH(W) and RH. □

Remark 21.6 (Exact signature of a repeated imaginary part). Under DBSRH(W), a repeated imaginary part at indices $k, k+1$ would force

$$\lim_{X\to\infty} \left[\varepsilon_k^{\mathrm{RvM}}(X;W) - \varepsilon_{k+1}^{\mathrm{RvM}}(X;W)\right] = h_k^{\mathrm{RvM}}. \tag{21.10}$$

Thus the entire positive RvM spacing would be consumed by a *difference* of arithmetic displacements. This is a precise target for a prime-based non-collapse estimate; it is not, without such an estimate, a contradiction. Mean-square bounds such as (2.4) or their taucm analogue control an *average over indices in height windows*; they do not exclude a single exceptional pair at a fixed k, even if the average RMS is comparable to the RvM mean spacing.

21.4 A derivative-based sufficient condition, and why it is too strong

The local Lipschitz inequality of Proposition 16.3 illustrates one way to obtain a cutoff-independent spacing, but its required uniform derivative control is not available for the Euler phase. The following sharpened criterion also identifies an *obstruction to using that particular proof strategy.*

Proposition 21.7 (Sobolev sufficient condition on a fixed index band)**.** *Fix a sufficiently large k and let $J_k = [a_k, b_{k+1}]$, with the barriers from Theorem 12.1. If, for some $p > 1$ and $K_k \in (0,\infty)$,*

$$\sup_{X\geq X_k} \left\|(F_X^W)'\right\|_{L^p(J_k)} \leq K_k, \tag{21.11}$$

then, for every $X \geq X_k$,

$$\boxed{\Delta_k(X;W) \geq K_k^{-p/(p-1)} > 0.} \tag{21.12}$$

Proof. For each cutoff the barrier values yield a passage within J_k from the level $k - \frac{1}{2}$ to $k + \frac{1}{2}$. Choose $s_0 < s_1$ in J_k by the extremal construction in the proof of Proposition 16.3, so that the passage occurs without leaving the open level band on (s_0, s_1). It is part of the set counted by Δ_k, and absolute continuity and Hölder's inequality give

$$1 \leq \int_{s_0}^{s_1} |(F_X^W)'(t)|\, dt \leq K_k(s_1 - s_0)^{1-1/p} \leq K_k \Delta_k(X;W)^{1-1/p}.$$

Rearranging proves the claimed bound. □

Remark 21.8 (Why uniform derivatives are not the proposed solution). The hypothesis (21.11) is much stronger than the occupation condition of Proposition 12.3 and is *not* a consequence of the mean-value bounds of this note. Moreover, if an arithmetic family converges pointwise away from the zero heights to a function with jumps on J_k, a uniform $W^{1,p}(J_k)$ bound with $p > 1$, together with the fixed MCOM amplitude bounds, is impossible: the one-dimensional Morrey inequality would make the family equicontinuous, and the uniform bound $|F_X^W| \leq |N_0| + H$ on

J_k then makes it compact in $C(J_k)$ by Arzelà–Ascoli, whereas a continuous subsequential limit cannot agree on both sides of a jump with the discontinuous pointwise limit. Thus controlling all derivatives uniformly in the cutoff is not the plausible path. A more local prime-only occupation or correlated *increment* estimate is required; the obstruction is to this strong regularity strategy, not to the taucm(s) programme itself.

21.5 Why the numerically accurate pair centers do not close the proof

For clarity, define

$$c_k(X;W) := \tfrac{1}{2}(\tau_k^{\mathrm{Cms}}(X;W) + \tau_{k+1}^{\mathrm{Cms}}(X;W)), \qquad s_k(X;W) := \Delta_k(X;W). \tag{21.13}$$

Knowledge of c_k and s_k is equivalent to knowledge of the ordered pair, but an accurate center does not bound s_k from below: $c_k \pm s_k/2$ may approach the same limit with $s_k \downarrow 0$. The accurate numerical centers in Section 20 are therefore valuable for choosing sampling windows for the untruncated Euler phase, not a proof of separation. The quadratic inverse theorem Theorem 19.1 and the prime-only occupation inequality (12.5) are two possible ways to attack the missing arithmetic estimate. Neither currently provides the uniform error control for all indices or the fixed-index limit $X \to \infty$.

Precise research target. For every fixed k, find a statement proved directly for the weighted prime phase P_X^W and the MCOM-truncated sublevel measures that yields one of the equivalent inequalities (21.4)–(21.5), preferably through positive occupation of the level band or a controlled correlation of consecutive arithmetic displacements. Such a theorem must allow the phase derivative to grow with X, respect the full weight support $n \le X^2$, and not assume information about β_k or the absence of repeated γ_k. Together with an independently proved DBSRH(W) it would give RH and simplicity; without DBSRH, it does not by itself locate the genuine imaginary parts.

22 Arithmetic scope and source attribution

This section integrates the deductions developed in the present research conversation with the author's public working notes at commit `38aac7c` [17, 18]. The sources are not identical in status. The earlier taucs τ_k^{C} and taucm τ_k^{Cm} use the author's interpolated *first-prime Euler phase* Φ°; the taucms $\tau_k^{\mathrm{Cms}}(X;W)$ use a *different, compensated, weighted* prime-power phase. In particular the weight $W_{1/4}$ is a chosen external smooth weight and is *not* presented as an invention of the author. The elementary sublevel, projection, Fourier, complex-analysis and moment inequalities used below are mathematical techniques, not separately claimed discoveries. The possible novelty lies in the combination with the author's ordinal objects and the specific MCOM coefficient. None of the present deductions is a proof of RH.

Three different assertions must remain separate:

(i) strict positivity of $\tau_{k+1}^{\mathrm{Cms}}(X;W) - \tau_k^{\mathrm{Cms}}(X;W)$ for each finite X;

(ii) the noncollapse condition $\limsup_{X\to\infty}[\tau_{k+1}^{\mathrm{Cms}}(X;W) - \tau_k^{\mathrm{Cms}}(X;W)] > 0$ for a fixed index;

(iii) convergence of the *same* sublevel points to the corresponding imaginary parts of zeros (DBSRH).

For every sufficiently large index the first follows from the continuity and coercive MCOM barriers; for the initial indices it requires checking that the corresponding two levels are attained. The second plus the third, for all indices, would imply RH and simplicity; none follows from a bound for $S(t)$ alone.

23 The genuine MCOM margin, and the distinction from moment inputs

Write $A_{\mathrm{M}} = 0.08382$, $C_{\mathrm{H}} = 0.0748596142711$ and keep the *fixed, effective* truncation H of (3.2); the asymptotic coefficient is not silently substituted into that definition. For t outside zero heights write $G(t) = N(t) - N_0(t)$, with the right-continuous convention at the heights.

Proposition 23.1 (MCOM interior margin)**.** *Assume the two inequalities of the author's MCOM manuscript,* (3.1) *and* (3.9)*. For every fixed* $0 < \eta < A_{\mathrm{M}} - C_{\mathrm{H}}$ *there exists* T_η *such that*

$$H(t) - |G(t)| \geq \eta \log t \qquad (t \geq T_\eta) \tag{23.1}$$

away from zero heights and, by right limits, at the heights.

Proof. Choose c strictly between C_{H} and $A_{\mathrm{M}} - \eta$. The asserted limsup implies $|S(t)| \leq c \log t$ eventually. Since $G = S + R$ and $R(t) = O(t^{-1})$, choose a slightly larger intermediate coefficient, still below $A_{\mathrm{M}} - \eta$, to absorb $R(t)$. The 5.5 in H is nonnegative, and (23.1) follows. □

The margin is a mathematically concrete use of the *two* MCOM estimates. It is not, by itself, a new Selberg–Fujii moment bound. The inequality $|G| \leq H$ also supplies the exact projection identity, for any real prime phase P,

$$|P - G| - |\operatorname{clip}_H(P) - G| = (|P| - H)_+. \tag{23.2}$$

The right side is zero on nonsaturated points and equals the distance removed by the truncation on saturated points. The explicit MCOM bound makes the target G lie inside the interval $[-H, H]$; the asymptotic refinement gives the stronger margin (23.1). Neither property implies pointwise convergence of the prime phases.

Proposition 23.2 (Crossing budget localized by the MCOM barriers)**.** *Let* F_X^W *be the actual taucms function from* (5.5)*,* P_X^W *its* untruncated *phase, and* $\lambda_k = k - \frac{1}{2}$*. For every* k *sufficiently large, let* $a_k < b_k$ *be the MCOM barriers specified in Theorem 12.1 by* $N_0(a_k) + H(a_k) = \lambda_k$ *and* $N_0(b_k) - H(b_k) = \lambda_k$*. Then, for every* $X \geq 2$ *and* $W \in \mathcal{W}$*,*

$$|\tau_k^{\mathrm{Cms}}(X; W) - \gamma_k| \leq \operatorname{meas}\{t \in [a_k, b_k] : |P_X^W(t) - G(t)| \geq \tfrac{1}{2}\}. \tag{23.3}$$

The same statement holds for τ_k^{Cm} *with* P_X^W *replaced by* Φ°*.*

Proof. The sublevel indicator of $N(t)$ at λ_k equals $\mathbf{1}_{\{t < \gamma_k\}}$ a.e., even for repeated imaginary parts, because the zeros are counted with multiplicity. For any integer n and real x, differing indicators of $x < k - \frac{1}{2}$ and $n < k - \frac{1}{2}$ imply $|x - n| \geq \frac{1}{2}$. The MCOM bound $|G| \leq H$ and projection contractivity yield $|F_X^W - N| \leq |P_X^W - G|$. Finally, the MCOM barrier inequalities place both sublevel boundaries in $[a_k, b_k]$ up to null sets; outside this interval the two indicators agree. Integration proves the claim. The constants here are *not* claimed to be sharp enough to force the right side to vanish. □

Remark 23.3 (What the 1/2 crossing does and does not do). The asymptotic MCOM coefficient yields $\limsup_{k\to\infty} |\gamma_k - \gamma_k^{\mathrm{RvM}}| \leq 2\pi C_{\mathrm{H}} < 1/2$ under the MCOM assertion. This strict numerical threshold is not supplied by the earlier coefficients discussed in the source notes. It is a proximity statement in *physical units of the imaginary part*; it is not a bound by half the RvM spacing, which is $\asymp 1/\log k$ and tends to zero. The equality "limsup $= 2\pi\mathfrak{a}$" quoted in the exploratory notes is *not* used: the necessary consequence here is the inequality $\leq$. Neither this threshold nor (23.3) excludes repeated imaginary parts on its own.

24 Quantitative ordinal transport under the cited fourth moment

The following is an audited formulation of the transport statement first written in the conversation and saved in [18]. It does not improve the fourth moment of Selberg or Fujii. It improves the *consequence for the ordinal points* relative to what the moment statement alone says.

Definition 24.1 (Auxiliary untruncated sublevels)**.** Define, solely for comparison,

$$u_k^\circ := \operatorname{meas}\{t > 0 : N_0(t) + \Phi^\circ(t) < k - \tfrac{1}{2}\}. \tag{24.1}$$

These are neither the original taucs nor the taucm(s). Let $\mathcal{K}_T = \{k : T \le \gamma_k^{\mathrm{RvM}} \le 2T\}$.

Theorem 24.2 (Transport of an MCOM-clipped Euler phase)**.** *Assume the untruncated fourth-moment input* (3.6) *from the author's IJNT manuscript and both MCOM estimates* (3.1) *and* (3.9)*. Then, as* $T \to \infty$*,*

$$\sum_{k\in\mathcal{K}_T} |\tau_k^{\mathrm{Cm}} - u_k^\circ| \ll \frac{T}{(\log T)^3}, \tag{24.2}$$

$$\frac{1}{\#\mathcal{K}_T} \sum_{k\in\mathcal{K}_T} |\tau_k^{\mathrm{Cm}} - u_k^\circ| \ll \frac{1}{(\log T)^4}. \tag{24.3}$$

For fixed $\epsilon > 0$*, the fraction of indices satisfying* $|\tau_k^{\mathrm{Cm}} - u_k^\circ| > \epsilon(\gamma_{k+1}^{\mathrm{RvM}} - \gamma_k^{\mathrm{RvM}})$ *is* $O_\epsilon((\log T)^{-3})$*. The analogous total* L^1 *bound for the difference of consecutive spacings holds with both indices in* $\mathcal{K}_T$*.*

Proof. Let $P(t) = \Phi^\circ(t)$ and $D(t) = (|P(t)| - H(t))_+$. By Proposition 23.1, on $[T/2, 3T]$, for T sufficiently large, the set $E_T = \{D > 0\}$ satisfies $|P - G| \ge D + c\log T$, with $c > 0$ fixed. Thus

$$\operatorname{meas}(E_T) \ll \frac{1}{(\log T)^4}\int_{T/2}^{3T} |P-G|^4 dt \ll \frac{T}{(\log T)^4}, \quad \int_{T/2}^{3T} D\,dt \ll \frac{T}{(\log T)^3}. \tag{24.4}$$

The last inequality follows from $D \le |P-G|^4/(c\log T)^3$ on E_T. The cited IJNT fourth moment applies on the requisite finite collection of dyadic blocks.

To justify localization independently of clipping, expand the finite Euler logarithms and sum the elementary bounds: $|\Phi^\circ(t)| \ll (1+t)^{1/4}\sqrt{\log(3+t)}$. Consequently $N_0 + P$ and $N_0 + \mathrm{clip}_H P$ are both below every level with $k \in \mathcal{K}_T$ when $t \le T/2$ and above it when $t \ge 3T$, for all large T. The initial compact interval is absorbed by the same asymptotic comparison.

For fixed t the sublevel indicators for these two functions can disagree only at levels $k - \frac{1}{2}$ within an interval of length $D(t)$. If $D(t) > 0$, that interval contains at most $D(t) + 1$ such half-integer levels. Therefore

$$\sum_{k\in\mathcal{K}_T} |\tau_k^{\mathrm{Cm}} - u_k^\circ| \le \int_{T/2}^{3T} [D(t) + \mathbf{1}_{E_T}(t)]\,dt \ll T(\log T)^{-3}.$$

The cardinality of $\mathcal{K}_T$ is $\asymp T\log T$. The pointwise triangle inequality for two consecutive spacings proves their L^1 comparison; Markov's inequality and $\gamma_{k+1}^{\mathrm{RvM}} - \gamma_k^{\mathrm{RvM}} \asymp (\log T)^{-1}$ give the stated exceptional proportion. □

Corollary 24.3 (Uniform transport over the proved weighted cutoff range)**.** *Assume the uniform fourth-moment theorem of Section 7, with its stated Fujii–IJNT input, and the MCOM estimates. For a fixed* $0 < \epsilon < 1/4$ *define the* auxiliary *untruncated points*

$$u_k^\circ(X; W) = \operatorname{meas}\{t > 0 : N_0(t) + P_X^W(t) < k - \tfrac{1}{2}\}.$$

Uniformly for $W \in \mathcal{W}$ and $T^{\epsilon} \le X \le T^{1/4}$,

$$\sum_{k \in \mathcal{K}_T} |\tau_k^{\mathrm{Cms}}(X; W) - u_k^{\circ}(X; W)| \ll_{\epsilon} \frac{T}{(\log T)^3}. \tag{24.5}$$

Proof. The uniform moment input replaces the fixed Φ° input in (24.4). The finite weighted Euler sum has the absolute bound $O(X)$, while the pole compensator has the bound $O(X/t)$ for $t \ge 1$; at $0 < t < 1$ it has the trivial bound $O(X)$ as well. For $X \le T^{1/4}$ these bounds are $o(T \log T)$, enough to localize the relevant sublevel indicators to $[T/2, 3T]$. The preceding level-counting argument then gives the bound uniformly. □

Remark 24.4 (No improvement of Selberg–Fujii itself). Neither (24.2) nor (24.5) improves a fourth-moment exponent, constant or allowed prime cutoff in Selberg–Fujii. Further, the *rate* $T/(\log T)^3$ follows from a positive logarithmic margin and the cited fourth moment; once those inputs are assumed, that elementary step does not uniquely require the specific decimal 0.08382. The particular MCOM estimates are what justify the margin and the covered projection for the *specified* taucm construction. The auxiliary points u_k° are *not* the true imaginary parts, and these results do not imply noncollapse for a fixed k as $X \to \infty$.

Remark 24.5 (The asymptotic estimate (3.9) is not needed for the rate). The conclusions of Theorem 24.2 and Corollary 24.3 hold with the explicit bound (3.1) alone, provided Selberg's unconditional moment estimate $\int_T^{3T} |S(t)|^6\, dt \ll T(\log\log T)^3$ [9] is added. Split $[T/2, 3T]$ into $B_1 = \{|G| \le H/2\}$ and $B_2 = \{|G| > H/2\}$. On $E_T \cap B_1$ one has $|P - G| \ge D + H/2 \ge D + \frac{1}{2}A_{\mathrm{M}} \log(T/2)$, and the Markov step of the proof applies verbatim with $c = \frac{1}{2}A_{\mathrm{M}}$. On B_2, $|S| \ge H/2 - |R| \ge \frac{1}{2}A_{\mathrm{M}} \log T + 2$ for large T, so $\mathrm{meas}(B_2) \ll T(\log\log T)^3 (\log T)^{-6}$. Since $|G| \le H$, $D \le |P - G|$ pointwise, and Hölder's inequality with the fourth moment gives $\int_{B_2} D\, dt \le \big(\int |P - G|^4\big)^{1/4} \mathrm{meas}(B_2)^{3/4} \ll T(\log\log T)^{9/4} (\log T)^{-9/2} \ll T(\log T)^{-3}$, while $\mathrm{meas}(E_T \cap B_2) \le \mathrm{meas}(B_2) \ll T(\log T)^{-4}$. Thus the only property of the truncation used by the transport rate is $H \asymp \log t$ together with the covering $|G| \le H$; the asymptotic coefficient C_{H} removes the factor $(\log\log T)^{O(1)}$ from the exceptional set but does not change the order $T(\log T)^{-3}$.

25 A prime-by-prime inequality and its exact limitation

For the original Euler phase (2.1), adding a prime p changes the finite phase by

$$\phi_p(t) = -\frac{1}{\pi} \arg(1 - p^{-1/2 - it}), \qquad |\phi_p(t)| \le \frac{\arcsin(p^{-1/2})}{\pi} =: \alpha_p. \tag{25.1}$$

The bound is elementary geometry of the circle centered at 1 of radius $p^{-1/2}$. It is independent of MCOM; MCOM enters only when these phases are used with the fixed, covered truncation H. If an *auxiliary* phase at a fixed number of primes is called $B_m(t) = N_0(t) + \mathrm{clip}_{H(t)}\, \Phi_m(t)$, then $|B_{m+1} - B_m| \le \alpha_{p_{m+1}}$. Consequently

$$\begin{aligned} &\mathrm{meas}\{k - \tfrac{1}{2} \le B_{m+1} < k + \tfrac{1}{2}\} \\ &\quad \ge \mathrm{meas}\{k - \tfrac{1}{2} + \alpha_{p_{m+1}} \le B_m < k + \tfrac{1}{2} - \alpha_{p_{m+1}}\}. \end{aligned} \tag{25.2}$$

This is a finite-step interior-band inclusion, not persistence for an infinite sequence of prime cutoffs. The bounds $\alpha_p \sim 1/(\pi\sqrt{p})$ are not summable over primes. The structural saturation model of Section 4 also shows why pointwise clipping and diminishing step sizes cannot alone rule out a band collapsing in a limit. The compensated weighted family P_X^W is *not* the same as the discrete family Φ_m, and (25.2) is not claimed as a theorem about its increments.

26 A zero-free Euler transform: a conditional RH criterion

This alternative route refines the transform idea in [17]. It uses the *actual compensated weighted prime phase* P_X^W, not a new approximation or a relabeling of $S(t)$. The transform itself is elementary Fourier–Laplace analysis; only its application to this ordinal program is under discussion. This is an *alternative analytic route*, not an established prime-band persistence result, and not a proof of RH.

Fix $W \in \mathcal{W}$ once and for all. For $\Re s > 1/2$ put

$$k_s(t) = \frac{1}{(s-\frac{1}{2}-it)^2} + \frac{1}{(s-\frac{1}{2}+it)^2}, \qquad \mathcal{T}[f](s) = -\int_0^\infty f(t)\partial_t k_s(t)\,dt. \tag{26.1}$$

For every compact $K \Subset \{\Re s > 1/2\}$, $|\partial_t k_s(t)| \le C_K(1+t)^{-3}$, so the transform is holomorphic whenever $\int_0^\infty |f(t)|(1+t)^{-3}dt < \infty$.

Lemma 26.1 (Exact finite-cutoff Euler transform)**.** *For $X \ge 2$ and $\Re s > 1/2$,*

$$\mathcal{T}[P_X^W](s) = -\sum_{n\ge 2} W\Big(\frac{\log n}{\log X}\Big)\frac{\Lambda(n)\log n}{n^s} + \int_2^\infty W\Big(\frac{\log u}{\log X}\Big)\frac{\log u}{u^s}\,du. \tag{26.2}$$

Proof. The sum and integral defining P_X^W have compact support in their n, u variables for fixed X. For $a > 0$ and $\Re(s-\frac{1}{2}) > 0$, Fourier–Laplace inversion gives $\int_0^\infty k_s(t)\cos(at)dt = \pi a e^{-(s-\frac{1}{2})a}$. Apply integration by parts to each term of P_X^W; every sine term vanishes at $t = 0$ and its derivative contributes $-a\cos(at)/\pi$. Set $a = \log n$ or $\log u$. □

Theorem 26.2 (Weighted L^2 normality implies RH)**.** *Fix a weight $W \in \mathcal{W}$. Suppose that for some unbounded sequence of cuts $X_j \to \infty$,*

$$\sup_j \int_0^\infty \frac{|P_{X_j}^W(t)|^2}{(1+t)^3}\,dt < \infty. \tag{26.3}$$

Then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$. No Cauchy-in-measure hypothesis is needed.

Proof. Cauchy–Schwarz and the bound following (26.1) show that the holomorphic family $\{\mathcal{T}[P_{X_j}^W]\}_j$ is locally bounded on $\Re s > 1/2$. It is therefore normal by Montel's theorem. For $\Re s > 1$, dominated convergence in (26.2), using $0 \le W \le 1$ and $W(v) = 1$ for $v \le 1$, gives the pointwise limit

$$Q(s) := -\Big(\frac{\zeta'}{\zeta}\Big)'(s) + \int_2^\infty \frac{\log u}{u^s}\,du, \qquad \Re s > 1. \tag{26.4}$$

The integral equals $2^{1-s}\big(\log 2/(s-1) + (s-1)^{-2}\big)$; its double pole at $s = 1$ cancels the double pole of the first summand. Thus Q admits a meromorphic continuation to $\Re s > 1/2$, with a nonremovable double pole at any zero ρ in that half-plane. Any locally uniform subsequential limit provided by Montel is holomorphic on the whole half-plane and agrees with Q for $\Re s > 1$. Uniqueness of analytic continuation therefore forbids such a ρ. By the functional-equation symmetry $\rho \mapsto 1-\overline{\rho}$, the absence of zeros to the right of $1/2$ implies RH. □

Corollary 26.3 (MCOM excess criterion: an exact sufficient route)**.** *For the fixed effective MCOM truncation $H(t)$, let the* untruncated excess *be*

$$E_X(t) = \big(|P_X^W(t)| - H(t)\big)_+. \tag{26.5}$$

If for some $X_j \to \infty$

$$\sup_j \int_0^\infty \frac{E_{X_j}(t)^2}{(1+t)^3}\,dt < \infty, \tag{26.6}$$

then RH holds. Contrapositively, if RH is false, then for every fixed $W \in \mathcal{W}$,

$$\lim_{X\to\infty} \int_0^\infty \frac{\big(|P_X^W(t)| - H(t)\big)_+^2}{(1+t)^3}\, dt = +\infty. \tag{26.7}$$

Proof. Pointwise, $|P_X^W| \le H + E_X$. As $\int_0^\infty H(t)^2(1+t)^{-3}dt < \infty$, the triangle inequality in weighted L^2 converts (26.6) to (26.3). For the contrapositive, if the limit did not diverge, there would be an unbounded sequence X_j with bounded excess norm, contradicting the theorem. □

Remark 26.4 (How MCOM enters and what remains open). The last proof uses $H(t)$ as an explicit *scale*, but uses only $H \in L^2((1+t)^{-3}dt)$; its logical implication to RH is therefore not a new theorem that relies numerically on the coefficient 0.08382. To exploit the *sharpness* of MCOM one must actually establish (26.6) from the arithmetic coefficients or an MCOM-sensitive inequality. This has *not* been done. The normal-family technique is classical; the finite Euler identity (26.2) and the precise weighted-excess condition are recorded to identify a sharply formulated open task. There is no demonstrated improvement over Selberg–Fujii, nor is the converse "RH implies (26.6)" established.

27 Identification of the limit phase, and the first index

The criterion of Theorem 26.2 uses no convergence of the prime phases. If convergence in measure is added, the limit phase is identified exactly, and the first sublevel point of (5.5) is seen *not* to converge to γ_1. The deductions of this section were made in the research conversation of 2 November 2026; their external inputs are Malmstén's integral for $\log\Gamma$ [6, §12.31], the boundary formula for Cauchy integrals of L^2 functions [5, Thm. 93], the Hadamard product of ξ, and the explicit bound $|R(t)| \le 0.53/(\pi t)$ already used in (3.2). Throughout, $N_\vartheta = \vartheta/\pi + 1$, so that $N = N_\vartheta + S$ for every $t > 0$ by (1.2), $b := s - \frac12$, and $\psi = \Gamma'/\Gamma$.

Lemma 27.1 (Transforms of the smooth terms)**.** *Let $\Re s > \frac12$.*

(i) For $a > 0$, $\mathcal{T}[\sin(a\,\cdot)](s) = \pi a^2 e^{-ab}$; moreover $\mathcal{T}[1](s) = 2/b^2$ and $\mathcal{T}[t](s) = 0$.

(ii) $\mathcal{T}[N_\vartheta](s) = 2/b^2 - \frac14\psi'(s/2)$.

(iii) Put

$$J(t) = \int_0^{\log 2} e^{v/2}\frac{\sin(tv)}{v}\, dv, \qquad r_0(t) = -1 + \frac{1}{\pi}\arctan(2t) + \frac{1}{\pi}J(t). \tag{27.1}$$

Then r_0 is real-analytic, $r_0(0) = -1$, $J(t) > 0$ and $|r_0(t)| \le 1.46/t$ for $t > 0$, and

$$\mathcal{T}[r_0](s) = E(s) := \frac{1}{s^2} - \frac{2}{b^2} + \int_1^2 (\log u)\, u^{-s}\, du. \tag{27.2}$$

(iv) For $\Re s > 1$ the limit Q of (26.4) satisfies $Q(s) = -(\xi'/\xi)'(s) - E(s) - \mathcal{T}[N_\vartheta](s)$.

Proof. (i) Integrating by parts, $\mathcal{T}[\sin(a\,\cdot)](s) = a\int_0^\infty \cos(at)k_s(t)\,dt = \frac{a}{2}\int_{\mathbb{R}} e^{iat}k_s(t)\,dt$, since k_s is even. For $a > 0$ and $\Re b > 0$, residues give $\int_{\mathbb{R}} e^{iat}(b+it)^{-2}dt = 2\pi a e^{-ab}$ and $\int_{\mathbb{R}} e^{iat}(b-it)^{-2}dt = 0$. Next, $\mathcal{T}[1] = k_s(0) - k_s(\infty) = 2/b^2$, and $\mathcal{T}[t] = \int_0^\infty k_s\,dt = \frac12\int_{\mathbb{R}} k_s\,dt = 0$ because each of $(b \pm it)^{-2}$ integrates to zero over $\mathbb{R}$.

(ii) Malmstén's formula gives, for $\Re z > 0$, $\log\Gamma(z) = \int_0^\infty \big[(z-1)e^{-u} - \frac{e^{-u}-e^{-zu}}{1-e^{-u}}\big]\frac{du}{u}$. With $z = \frac14 + \frac{it}{2}$, $\Im\log\Gamma(z) = \int_0^\infty \big[\frac{t}{2}e^{-u} - \frac{e^{-u/4}\sin(tu/2)}{1-e^{-u}}\big]\frac{du}{u}$, whose integrand is $O(t + ut^3)$ as $u \to 0^+$; hence Fubini's theorem allows $\mathcal{T}$ to be applied under the integral. By (i), the term linear in t

contributes nothing and $\mathcal{T}[\sin(\frac{u}{2}\cdot)] = \pi\frac{u^2}{4}e^{-bu/2}$, so $\mathcal{T}[\Im\log\Gamma(\frac{1}{4}+\frac{i\cdot}{2})] = -\frac{\pi}{4}\int_0^\infty \frac{u\,e^{-su/2}}{1-e^{-u}}\,du = -\frac{\pi}{4}\psi'(s/2)$. Since $\vartheta(t) = \Im\log\Gamma(\frac{1}{4}+\frac{it}{2}) - \frac{t}{2}\log\pi$ and $\mathcal{T}[t] = 0$, (ii) follows.

(iii) From $\arctan(2t) = \int_0^\infty e^{-v/2}\sin(tv)\,dv/v$ and (i), $\mathcal{T}[\pi^{-1}\arctan(2\,\cdot)] = \int_0^\infty ve^{-sv}dv = s^{-2}$ and $\mathcal{T}[\pi^{-1}J] = \int_0^{\log 2} ve^{(1-s)v}dv = \int_1^2 (\log u)u^{-s}du$; with $\mathcal{T}[-1] = -2/b^2$ this is (27.2). Analyticity and $r_0(0) = -1$ are clear. For the bounds write $J(t) = \mathrm{Si}(t\log 2) + \int_0^{\log 2}\varphi(v)\sin(tv)\,dv$ with $\varphi(v) = (e^{v/2}-1)/v$ increasing from $\frac{1}{2}$ to 0.598. Integration by parts gives $|\int_0^{\log 2}\varphi\sin(tv)\,dv| \le 2\varphi(\log 2)/t \le 1.2/t$, and also $\le \varphi(\log 2)\,t(\log 2)^2/2 \le 0.144\,t$. For $t \le 2.9$, $\mathrm{Si}(t\log 2) \ge 0.79\,t\log 2$ by concavity of Si on $[0,\pi]$, so $J(t) \ge 0.40\,t > 0$; for $t \ge 2.9$, $\mathrm{Si}(t\log 2) \ge \mathrm{Si}(2\pi) > 1.41$ and $1.2/t \le 0.42$, so $J(t) > 0$. Finally $r_0 = \pi^{-1}[J - \frac{\pi}{2} - \arctan\frac{1}{2t}]$ and $|\mathrm{Si}(x) - \frac{\pi}{2}| \le 2/x$ give $|r_0(t)| \le \pi^{-1}(2/\log 2 + 1.2 + \frac{1}{2})/t \le 1.46/t$.

(iv) From $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, $(\xi'/\xi)' = (\zeta'/\zeta)' - s^{-2} - (s-1)^{-2} + \frac{1}{4}\psi'(s/2)$. Using (ii), (iii) and $\int_1^\infty(\log u)u^{-s}du = (s-1)^{-2}$,

$$-(\xi'/\xi)' - E - \mathcal{T}[N_\vartheta] = -(\zeta'/\zeta)' + \int_1^\infty (\log u)u^{-s}du - \int_1^2 (\log u)u^{-s}du = Q(s). \qquad \square$$

Lemma 27.2 (Uniqueness for the transform)**.** *Let g be measurable on $(0,\infty)$ with $\int_0^\infty |g(t)|^2(1+t)^{-3}dt < \infty$ and $\mathcal{T}[g](s) = 0$ for all $\Re s > \frac{1}{2}$. Then $g = 0$ almost everywhere.*

Proof. Let g_o be the odd extension of g to $\mathbb{R}$. Since $\partial_t k_s$ is odd, $\mathcal{T}[g](s) = -\frac{1}{2}\int_{\mathbb{R}} g_o\,\partial_t k_s\,dt$, and $\partial_t k_s(t) = 2(t+ib)^{-3} + 2(t-ib)^{-3}$. Put $C(z) = \int_{\mathbb{R}} g_o(t)(t-z)^{-3}dt$ for $z \notin \mathbb{R}$ (absolutely convergent and holomorphic). Oddness gives $C(-z) = C(z)$, so $\mathcal{T}[g](s) = -2C(ib)$, and the hypothesis says $C \equiv 0$ on $\{\Im z > 0\}$, hence on $\{\Im z < 0\}$ as well. Let $C_1(z) = \int_{\mathbb{R}} g_o(t)\big[(t-z)^{-1} - (t-i)^{-1} - (z-i)(t-i)^{-2}\big]dt$; the bracket is $O(|t|^{-3})$, C_1 is holomorphic off $\mathbb{R}$ and $C_1'' = 2C \equiv 0$, so C_1 is a quadratic polynomial $p_\pm$ on each half-plane. The bracket equals $(z-i)^2(t-z)^{-1}(t-i)^{-2}$, so $C_1(z) = (z-i)^2\int_{\mathbb{R}} w(t)(t-z)^{-1}dt$ with $w(t) = g_o(t)(t-i)^{-2} \in L^2(\mathbb{R})$. By the boundary formula for Cauchy integrals of L^2 functions [5, Thm. 93], $2\pi i\,w(x) = \lim_{y\to 0^+}[C_1(x+iy) - C_1(x-iy)](x-i)^{-2} = (p_+(x) - p_-(x))(x-i)^{-2}$ for almost every x. Hence g_o agrees a.e. with a polynomial of degree at most two; being odd, $g_o(t) = ct$, and $\int_0^\infty c^2t^2(1+t)^{-3}dt < \infty$ forces $c = 0$. $\square$

Proposition 27.3 (Limit phase and the first index)**.** *Fix $W \in \mathcal{W}$. Suppose that a sequence $X_j \to \infty$ satisfies (26.3) and that $P^W_{X_j}$ converges in measure on every compact subset of $(0,\infty)$ to a function P_∞. Then:*

(a) RH holds and $P_\infty = S - r_0$ almost everywhere on $(0,\infty)$, with r_0 from (27.1);

(b) $|P_\infty(t)| < H(t)$ for every $t > 0$: the MCOM truncation is inactive in the limit, and $F_\infty := N_0 + \mathrm{clip}_H(P_\infty) = N - R - r_0$;

(c) $\lim_{j\to\infty}\tau_k^{\mathrm{Cms}}(X_j;W) = \gamma_k$ for every $k \ge 2$;

(d) $\lim_{j\to\infty}\tau_1^{\mathrm{Cms}}(X_j;W) = \gamma_1 - t_\sharp$, where

$$t_\sharp := \mathrm{meas}\{0 < t < \gamma_1 :\ R(t) + r_0(t) \le -\tfrac{1}{2}\} > 0. \tag{27.3}$$

Consequently DBSRH(W) *in the form (5.6) is incompatible with the two hypotheses. With the sublevel points measured from 2π, the minimum of N_0,*

$$\tau_k^\flat(X;W) := 2\pi + \mathrm{meas}\{t \ge 2\pi :\ F_X^W(t) < k - \tfrac{1}{2}\}, \tag{27.4}$$

one has $\lim_j \tau_k^\flat(X_j;W) = \gamma_k$ for every $k \ge 1$.

Proof. (a) RH is Theorem 26.2. By Fatou's lemma P_∞ satisfies the weighted L^2 bound; the bound gives uniform integrability on compact sets and, by Cauchy–Schwarz, uniformly small contributions of $(0,\delta)\cup(B,\infty)$ to $\int |P_{X_j}^W|(1+t)^{-3}dt$, so $\mathcal{T}[P_{X_j}^W](s) \to \mathcal{T}[P_\infty](s)$ for every $\Re s > \frac12$. Hence $\mathcal{T}[P_\infty]$ is the holomorphic continuation of Q obtained in the proof of Theorem 26.2. Integrating by parts, $\mathcal{T}[N](s) = \sum_k k_s(\gamma_k) = \sum_\rho (s-\rho)^{-2}$ under RH, which equals $-(\xi'/\xi)'(s)$ by the Hadamard product. By Lemma 27.1(iv), on $\Re s > \frac12$ (where both sides are holomorphic under RH), $\mathcal{T}[S - r_0] = \mathcal{T}[N] - \mathcal{T}[N_\vartheta] - \mathcal{T}[r_0] = -(\xi'/\xi)' - \mathcal{T}[N_\vartheta] - E = Q = \mathcal{T}[P_\infty]$. Lemma 27.2 applied to $g = P_\infty - S + r_0$, which satisfies the weighted L^2 bound since $|S| \ll \log(2+t)$ and r_0 is bounded, proves (a).

(b) For $0 < t < \gamma_1$ we have $N = 0$, so $S = -1 - \vartheta/\pi \in [-1, 0.13]$ because $\vartheta \geq -3.531$ on $(0,\infty)$; also $|J(t)| \leq t\int_0^{\log 2} e^{v/2}dv \leq 0.83\,t$, so for $t < 2$, $|r_0| \leq 1.53$ and $|S - r_0| \leq 2.6 < 5.5 \leq H(t)$. For $t \geq 2$, (3.1) (extended to zero heights by right limits) and $|r_0| \leq 0.73$ give $|S - r_0| \leq 0.08382\log t + 5.18 < H(t)$. Thus $\mathrm{clip}_H(P_\infty) = P_\infty$ and $F_\infty = N_0 + S - r_0 = N - R - r_0$.

(c), (d) Since $F_X^W \geq N_0 - H$, every sublevel set at level $\lambda_k = k - \frac12$ lies in $(0, B_k]$ with $B_k := \sup\{N_0 - H < \lambda_k\}$ independent of X. The truncation is 1-Lipschitz, so $F_{X_j}^W \to F_\infty$ in measure on compact subsets of $(0, B_k]$. On each interval between consecutive distinct zero heights, $F_\infty - \lambda_k$ is real-analytic and not constant (as $R + r_0 \to 0$ at infinity and $R + r_0 \to \frac18 - 1$ as $t \to 0^+$), so $\mathrm{meas}\{F_\infty = \lambda_k\} = 0$. Therefore, for each $\delta > 0$, $\mathrm{meas}(\{F_{X_j}^W < \lambda_k\} \cap [\delta, B_k]) \to \mathrm{meas}(\{F_\infty < \lambda_k\} \cap [\delta, B_k])$, while the part in $(0,\delta)$ has measure at most δ; letting $\delta \to 0$, $\lim_j \tau_k^{\mathrm{Cms}}(X_j; W) = \mathrm{meas}\{F_\infty < \lambda_k\}$. We compute this set. For $t \geq \gamma_1$, $|R| \leq 0.53/(\pi\gamma_1) < 0.012$ and $|r_0| \leq 1.46/\gamma_1 < 0.104$, so $|R + r_0| < \frac12$ and, N being an integer, $N - R - r_0 < k - \frac12$ holds exactly when $N \leq k - 1$, that is for $t < \gamma_k$. For $0 < t < \gamma_1$, $F_\infty = -R - r_0$; since $J > 0$, $-r_0 \leq 1 - \pi^{-1}\arctan(2t) < 1$, and $R > -\frac12$ there (for $t \geq 1$ by $|R| \leq 0.53/\pi$; for $t < 1$ by $\vartheta(t) \geq \vartheta(1) > -1.8$ and $N_0(t) \leq \frac78$, which give $R(t) > -0.45$), so $F_\infty < \frac32$. Hence $\{F_\infty < \lambda_k\} = (0, \gamma_k)$ up to a null set for every $k \geq 2$, proving (c), while $\{F_\infty < \frac12\} = \{t < \gamma_1 : R + r_0 > -\frac12\}$, whose measure is $\gamma_1 - t_\sharp$. Since $R(t) + r_0(t) \to -\frac78 < -\frac12$ as $t \to 0^+$ and both functions are continuous on $(0,\gamma_1)$, the set in (27.3) contains an interval $(0,\epsilon)$, so $t_\sharp > 0$. For (27.4): on $[2\pi, \gamma_1)$, $|R| \leq 0.53/(2\pi^2) < 0.03$ and $-r_0 \leq \pi^{-1}\arctan(1/(2t)) \leq 1/(4\pi^2)$, so $F_\infty < \frac12$ there, giving $\lim_j \tau_1^\flat = 2\pi + (\gamma_1 - 2\pi) = \gamma_1$; for $k \geq 2$ the interval $(0, 2\pi)$ lies inside $\{F_\infty < \lambda_k\}$, so $\tau_k^\flat$ and τ_k^{Cms} have the same limit. □

Remark 27.4 (Scope). The two hypotheses of Proposition 27.3 concern only the compensated weighted prime phase. The proposition does not prove RH; it shows that, in the only scenario in which the transform criterion applies, the limit of the prime phase is $S - r_0$ and not S, that the correction r_0 is invisible at every zero height above γ_1 (where $|R + r_0| < \frac12$), and that it displaces the first sublevel point by a positive amount $t_\sharp$. The numerical observation $P_X^W(0) = 0$ versus $N_0(0^+) = \frac78$ in Section 29 is the finite-cutoff trace of the same fact. The conditional statements of Corollary 21.5 and Section 30 remain as formulated; whenever DBSRH is assumed there, the first index should be read with (27.4).

Proposition 27.5 (Transform criterion for the normalized sublevel points)**.** *Fix $W \in \mathcal{W}$ and put, for $\Re s > \frac12$,*

$$\mathcal{K}_X(s) := \sum_{k\geq 1} k_s(\tau_k^\flat(X; W)), \tag{27.5}$$

which converges absolutely and locally uniformly, uniformly in $X \geq 2$, because $\tau_k^\flat \geq a_k$ for $k \geq k_0$ (Theorem 12.1) and $\sum_k a_k^{-2} < \infty$, while $2\pi \leq \tau_k^\flat \leq B_k$ for $k < k_0$. Let $\Sigma \subset \{\Re s > \frac12\}$ have an accumulation point in that half-plane. Then

$$\lim_{X\to\infty} \mathcal{K}_X(s) = -\Big(\frac{\xi'}{\xi}\Big)'(s) \text{ for every } s \in \Sigma \quad\Longleftrightarrow\quad \text{RH and } \lim_{X\to\infty} \tau_k^\flat(X; W) = \gamma_k \text{ for every } k \geq 1. \tag{27.6}$$

For $\Sigma \subset \{\Re s > 1\}$ the right side of the limit is the convergent prime expression $-(\zeta'/\zeta)'(s) - s^{-2} - (s-1)^{-2} + \frac{1}{4}\psi'(s/2)$, so the criterion compares the sublevel points with Euler, Γ and the pole only; for $s = \frac{1}{2} + d + iT$ the real and imaginary parts of $k_s(\tau)$ are the kernels of order $j = 2$ used in [2].

Proof. ($\Leftarrow$) Under RH, $-(\xi'/\xi)'(s) = \sum_\rho (s-\rho)^{-2} = \sum_k k_s(\gamma_k)$, and dominated convergence with the bound $|k_s(\tau)| \le C_s \min(1, \tau^{-2})$ gives the limit. ($\Rightarrow$) Let $X_n \to \infty$. As $\tau_k^\flat(X_n) \in [2\pi, B_k]$ is nondecreasing in k, a diagonal subsequence satisfies $\tau_k^\flat(X_{n_i}) \to \lambda_k$ for every k, with $2\pi \le \lambda_k \le \lambda_{k+1}$ and $\lambda_k \ge a_k$ for $k \ge k_0$. By dominated convergence $\mathcal{K}_{X_{n_i}} \to \mathcal{K}_\infty := \sum_k k_s(\lambda_k)$ locally uniformly on $\Re s > \frac{1}{2}$; the same series converges locally uniformly on $\mathbb{C}$ away from the points $\frac{1}{2} \pm i\lambda_k$, so $\mathcal{K}_\infty$ is meromorphic on $\mathbb{C}$ with double poles exactly at $\frac{1}{2} \pm i\lambda_k$, the leading coefficient at $\frac{1}{2} + i\mu$ being the number of k with $\lambda_k = \mu$. The hypothesis and the identity theorem for meromorphic functions give $\mathcal{K}_\infty = -(\xi'/\xi)'$ on $\Re s > \frac{1}{2}$; as $\mathcal{K}_\infty$ is holomorphic there, ξ has no zero in that half-plane, which is RH. Then $-(\xi'/\xi)' = \sum_k k_s(\gamma_k)$, and comparing the double poles and their coefficients of the two meromorphic functions shows that the multisets $\{\lambda_k\}$ and $\{\gamma_k\}$ coincide; both being enumerated in nondecreasing order, $\lambda_k = \gamma_k$ for every k. Since this holds along a subsequence of every sequence $X_n \to \infty$, the full limits exist. □

Remark 27.6. Proposition 27.5 is the precise content of the Laguerre–Pólya proposal in [17]: the finite-cutoff points are real by construction, so the only way the limit identity can fail is through a defect in the *positions* of the λ_k, never through a non-real limit. A repeated imaginary part is *not* excluded by the right side of (27.6): if $\gamma_k = \gamma_{k+1}$, RH still forces $\lambda_k = \lambda_{k+1}$ and the identity holds. Simplicity therefore has to come from the arithmetic noncollapse of Proposition 21.1, exactly as in Corollary 21.5.

28 A box-transform and a parity-sensitive multiplicity test

This section formalizes the third route in [17] *without* importing a zero-free claim or assuming RH. The identity is deterministic and holds for an arbitrary locally absolutely continuous real function F, including the actual taucm and taucms functions where their stated local regularity is available.

Theorem 28.1 (Finite-interval box identity, with the endpoint term)**.** *Let $0 < a < b$ not be imaginary parts of zeros. Suppose F is absolutely continuous on $[a, b]$. For a continuous test function g with a primitive A_g ($A_g' = g$), put $e_k(F) = k - \frac{1}{2} - F(\gamma_k)$ and $\Psi_g(z) = \int_{z-1/2}^{z+1/2} g(v)\,dv$. Then*

$$\sum_{a<\gamma_k<b} \Psi_g(e_k(F)) = \int_a^b g(N(t) - F(t))F'(t)\,dt + A_g(N(b) - F(b)) - A_g(N(a) - F(a)). \tag{28.1}$$

The sum counts imaginary parts of zeros with multiplicity.

Proof. On every interval between consecutive distinct zero heights, N is constant, and with $x = N - F$ we have $g(N-F)F' = -\frac{d}{dt}A_g(N-F)$ a.e. Sum these identities over the finitely many pieces in $[a, b]$. At an imaginary part with multiplicity m, the missing telescopic contribution is $A_g(x^- + m) - A_g(x^-)$, which equals $\sum_{i=0}^{m-1} \Psi_g(e_{k+i}(F))$ because the boxes $[e_{k+i} - 1/2, e_{k+i} + 1/2]$ tessellate $[x^-, x^- + m]$. The remaining terms are exactly the two endpoint values shown in (28.1). □

For $\xi \neq 0$, taking $g(v) = e^{i\xi v}$ yields the exact finite-interval Fourier form

$$\frac{2\sin(\xi/2)}{\xi} \sum_{a<\gamma_k<b} e^{i\xi e_k(F)} = \int_a^b e^{i\xi(N-F)} F'(t)\, dt + \frac{e^{i\xi(N(b)-F(b))} - e^{i\xi(N(a)-F(a))}}{i\xi}. \tag{28.2}$$

No "boundary terms are negligible" statement is assumed.

Proposition 28.2 (Parity bound for repeated heights)**.** *Let $\mathcal{I}$ be a union of complete repeated-height groups in the index list, and let $r(\mathcal{I})$ be the number of distinct imaginary parts represented. For any real c and any single-valued real function F,*

$$\#\mathcal{I} - r(\mathcal{I}) \leq \sum_{k\in\mathcal{I}} [1 - \cos\pi(e_k(F) - c)]. \tag{28.3}$$

Proof. For a group of multiplicity m we have $e_{k+i} = e_k + i$, $0 \leq i < m$; its right side is $m - \cos(\pi(e_k - c)) \sum_{i=0}^{m-1} (-1)^i \geq m - 1$. Sum over disjoint complete groups. □

The familiar relation $(-1)^{N(t)} = -\operatorname{sgn} Z(t)$ outside zeros, with the Hardy Z function normalized by $Z(0) < 0$, illustrates the information carried by the parity observable; it is *not* a new detection algorithm for off-line zeros. The box identity alone cannot make the right side of (28.3) vanish, and the sign of F' in (28.2) prevents discarding the oscillatory integral. This is therefore a separate exact potential tool, not an unconditional simplicity theorem.

29 Complex signed-pair kernel and proposed zero formula: status and limitations

In the external notes, a conditional complex continuation of the two-source kernel from Section 17 was proposed to separate a small real spacing from two mirror-symmetric zeros with the same imaginary part. It is important not to import this weighted model as if it were the author's *original* taucs or as if it supplied the full zero-explicit formula.

For the already defined entire kernel K and any complex h, $C(z,h) := K(z-h/2)+K(z+h/2)$ is entire and even in h. The Taylor expansion and its global remainder for real h are established in Section 17. Algebraically a real pair has $h^2 \geq 0$, a repeated imaginary part has $h^2 = 0$, and a formal transverse pair at $\gamma \mp id$ corresponds to $h^2 = -4L^2d^2$. The analytic continuation of the *model* is rigorous; the identification of its complex arguments with terms in the *actual weighted prime explicit formula* remains unproved. In particular, no infinite zero sum $\sum_\rho K(L(t-u_\rho))$ is asserted here as an unconditional identity. The contour deformation, branch constants, uniform convergence and contributions of the noncentral zeros must all be controlled first.

Proposition 29.1 (Signed central slope for an auxiliary weight)**.** *Let $q \geq 0$ be any nonzero, integrable weight on $[1,2]$, normalized by $\int_1^2 q(v)\, dv = 1$, and let $K_q(z) = \frac{1}{2} + \pi^{-1} \int_1^2 q(v) \operatorname{Si}(vz)\, dv$. For the* model *$C_q(z,h) = K_q(z - h/2) + K_q(z + h/2)$, its derivative at the pair center is*

$$\mathscr{S}_q(h) = \frac{2}{\pi} \int_1^2 q(v) \frac{\sin(vh/2)}{h/2}\, dv \quad (\mathscr{S}_q(0) = \tfrac{2}{\pi} \int_1^2 vq(v)\, dv).$$

For every real $h \neq 0$, $\mathscr{S}_q(h) < \mathscr{S}_q(0)$; for every real $d > 0$, $\mathscr{S}_q(2id) > \mathscr{S}_q(0)$. These comparisons classify the model *parameter with a sign; they are not a criterion about actual off-line zeta zeros without a validated zero-explicit formula.*

Proof. For real $x \neq 0$, $\sin x/x < 1$; for real $x \neq 0$, $\sinh x/x > 1$. Substitute into the integral, with $q \geq 0$. □

The symmetry $K_q(-z) = 1 - K_q(z)$ makes the derivative of C_q at the center stationary under a first-order shift of the sampling center, a useful *model* fact; the uncontrolled influence of neighboring contributions remains. The special weight $q(v) = v/2 + 1/4$ used in earlier numerical sections is one *borrowed choice* and is not needed for this auxiliary proposition.

The inverse estimator from Section 19 measures a signed algebraic quadratic parameter only if its sampling points and center are independently specified and the projected arithmetic residual is rigorously bounded. No such prime-only bound is currently known. The small-gap numerical experiments cannot supply one. For these reasons the signed-pair model is recorded as an *auxiliary exploratory route*, not as the main proposed RH mechanism.

A separate Laguerre–Pólya proposal in the public notes forms an even entire product from the real τ_k^{Cms}. Reality of finite-cutoff sublevel points and normality of such products cannot by themselves exclude a multiple real zero in a limit, so this is not an independent RH route. Any product-normality assertion also requires its stated uniform tail bounds and control of the initial factors.

The public notes also flag a lower-endpoint defect for the naive "DBSRH for every $k \geq 1$" statement: the Euler phase satisfies $P_X^W(0) = 0$, whereas the fixed main term extends to $N_0(0^+) = 7/8 > 1/2$. Proposition 27.3 makes this precise: under the hypotheses of the transform criterion together with convergence in measure, τ_1^{Cms} converges to $\gamma_1 - t_\sharp$ with $t_\sharp > 0$, while the points $\tau_k^\flat = 2\pi + \mathrm{meas}\{t \geq 2\pi : F_X^W(t) < k - \frac{1}{2}\}$ of (27.4) converge to γ_k for every $k \geq 1$. Finite-cutoff strictness of $\tau_k^\flat$ in the initial indices is not established here. We therefore retain the original definitions for existing theorems and do not silently replace DBSRH(W) by its corrected variant.

30 Open tasks after merging the research routes

The routes now have the following precise mathematical status.

(1) **Prime-band arithmetic persistence (open).** Prove $\limsup_{X\to\infty} \Delta_k(X; W) > 0$ for each k from the *compensated* prime-power phase, not merely from the shrinking single-prime perturbations or a saturating auxiliary model. Under the unproved DBSRH hypothesis this would imply RH and simplicity.

(2) **MCOM-critical local inequalities (open).** Seek a bound in which the actual constants $A_{\mathrm{M}} = 0.08382$ or $C_{\mathrm{H}} < 1/(4\pi)$ cross a threshold *derived from the ordinal equations*. The proximity $2\pi C_{\mathrm{H}} < 1/2$ is not itself such a theorem.

(3) **Compensated excess (open).** Prove (26.6) for an unbounded sequence of cutoffs, or a sufficiently strong uniform prime-band estimate implying it. By Corollary 26.3, this would imply RH, without needing a separate Cauchy-in-measure statement. It is a one-way sufficient criterion, *not* an established bound.

(4) **Parity and local pair residual (open).** Exploit the exact Fourier-box identity (28.2) or bound the residual in the signed-pair model without assuming off-line zero control. The current identities alone do not yield the required positivity or vanishing.

(5) **Endpoint and analytic inputs (open).** Verify the weighted zero-explicit formula including branch conventions and the exact IJNT/Fujii ranges (the first-index normalization is settled, under the hypotheses of Proposition 27.3, by (27.4)); no claims about $X \to \infty$ at fixed height follow from moment estimates established only for $X \leq T^{1/4}$.

31 Research priorities and exclusions

The current research problem is the *arithmetic structure of the prime-corrected sublevel constructions*; a stand-alone equivalence with RH via the Hadamard product, ξ'/ξ, a Littlewood integral or another classical reformulation is *not* proposed as a route. Traditional analytic techniques remain admissible when they yield new bounds on the prime phases, local occupation tests, their sublevel measures, or the evaluation defect (13.1).

1. Establish the *actual* weighted explicit formula for P_X^W, retaining the compensation at $u = 2$, with an error uniform in the range needed for $X \to \infty$ at fixed height. In particular, a hypothetical off-line zero cannot be treated as an isolated maximal-β zero unless that property is proved.

2. Prove a prime-only remainder estimate for the spatial-filter inverse in Theorem 19.1, or derive noncollapse via the local occupation quantities in Proposition 12.3, or obtain a stronger direct arithmetic bound on $\Delta_k(X; W)$. The separation of the γ_k^{RvM} is a structural input but not a substitute for this estimate.

3. Derive an unconditional second-order bound for $\sum e_k(F)^2$ on growing index windows for a specified prime budget. The numerical values of $1 - 4\eta$ do not prove an improved asymptotic proportion of zeros on the critical line.

4. Close the explicit $j = 1$ budget at $\theta = 1/2$ for the actual chosen family, including the B_T term and all endpoint corrections; the reported 82.6% is a numerical conditional estimate, not a theorem.

5. Treat moments of growing order separately from fixed-order Selberg–Fujii estimates. In particular, numerical successes for $2j \le 14$ at finite heights give no uniform estimate as $T \to \infty$.

6. Do not substitute the model $C(z, Lg)$ for an unconditional zero decomposition of the Euler phase. When differentiating in L, retain the moving-evaluation correction (18.6), and use the untruncated phase for derivatives.

7. Study the *consecutive difference* of RvM-relative arithmetic displacements in (21.3); do not infer its strict upper bound from individual MCOM proximity or from window-averaged mean-square estimates. The coefficient C_{H} belongs to a redefined asymptotic truncation, not to the present fixed H.

8. Do not conflate the three different limits: $X \to \infty$ with k fixed (DBSRH), $T \to \infty$ with $X = X(T)$ (mean-value results), and $j \to \infty$ (strict cell criteria).

Related cell-departure draft. The separate source `teorema_salida_celda_v1.tex` provides conditional, near-neighbor sufficient conditions for leaving a prescribed index interval. Its proposed explicit formula is written under RH and contains steps marked as sketches. It therefore cannot serve as an unconditional bridge from the finite prime phases to RH. Its local kernel analysis can nevertheless motivate a direct study of the occupation sets in Proposition 12.3, with all errors and quantifiers retained.

Disputed bridges that are not established. The implication RH ⇒ DBSRH in the separate DBSRH draft relies on a weighted explicit-formula argument that is not fully audited here. The unbounded-X uniform fourth moment, the isolated-maximum-free saturation assertion, and an L^4 Cauchy bound for every $\sqrt{T} \le X < X'$ are all *open*; no statement that "only the diagonal

term of Montgomery–Vaughan remains" is justified. The estimate in Theorem 7.3 uses the weighted support X^2, so the IJNT prime-index cutoff and $X \asymp \sqrt{T}$ are not interchangeable. A classical identity for the evaluation defects is an exact tool for studying the proposed constructions, not an independent established route to RH.

32 Conclusions and theorem-status table

Statement	Status	What the proof uses
MCOM-margin and localized crossing budget	Conditional on MCOM manuscript	MCOM effective and asymptotic constants; sublevel comparison, no RH.
Quantitative ordinal transport $O(T/(\log T)^3)$	Conditional on IJNT/Fujii input	MCOM interior margin plus cited fourth moment; does not improve Selberg–Fujii moments.
Exact finite Euler–pole transform	Unconditional identity	Fourier–Laplace inversion of actual weighted prime phase.
Weighted-L^2 normal-family criterion for RH	Proved implication; hypothesis open	Montel, Euler–pole transform; no measure convergence assumption.
Fourier-box identity and parity test	Exact	Finite-interval integration by parts and endpoint terms; no simplicity claim.
Limit phase $P_\infty = S - r_0$ and first-index defect $t_\sharp > 0$	Conditional on weighted-L^2 bound and convergence in measure	Transforms of N_ϑ and r_0, Hadamard product, uniqueness via Cauchy integrals; MCOM bound for inactivity of the truncation.
Transform criterion for $\tau_k^\flat$	Proved equivalence; hypothesis open	Normality from the MCOM barriers, identity theorem, pole comparison; RH together with convergence of all $\tau_k^\flat$.
Signed transverse two-source identification	Exploratory only	Analytic continuation of a model; missing weighted zero-explicit formula.
Failure of eventual neighboring-zero containment for $h = \log\log t$	Unconditional	Tsang's $\Omega_\pm$ theorem and an elementary counting argument.
Fourth moment and mean-square approximation for taucm	Unconditional subject to the cited IJNT inputs	MCOM bound, projection, IJNT fourth moment, gaps, and sublevel transfer.
Structural saturation model	Unconditional	Explicit continuous functions; no Euler phase.
Saturation incompatible with a uniform $O(T)$ fourth moment	Unconditional for the model	Weak equidistribution and the identity $\frac{1}{2}(H-G)^4 + \frac{1}{2}(-H-G)^4$.
Failure for $X = T^{o(1)}$	Unconditional subject to the Fujii input	Selberg approximation and Montgomery–Vaughan.
Uniform bound for $T^\varepsilon \le X \le T^{1/4}$	Unconditional subject to the Fujii input	Weighted prime-power decomposition and MCOM truncation.
Smoothed zero-sum bound for $\log X \ge \varepsilon \log T$	Unconditional subject to the Fujii increment input	Zero-count increments and Minkowski.
Improved range $X \le T^{1/4}(\log T)^{3/2}$ for two weights	Unconditional with Fujii input	Endpoint decay of the weights and Montgomery–Vaughan.
Exact finite-cutoff margin and Lipschitz spacing	Unconditional	Triangle identity and local Lipschitz regularity; not a uniform lower bound in X.
Correlated RvM displacement and fixed-H proximity	Exact / unconditional	RvM mean-value theorem, the existing MCOM barrier, and difference of consecutive shifts.

Statement	Status	What the proof uses
Noncollapse criterion under DBSRH	Conditional	Strict one-sided bound on the correlated shifts; requires an additional prime-only proof.
Uniform local derivative sufficient condition	Conditional	A uniform $W^{1,p}$ bound; generally incompatible with a discontinuous pointwise phase limit.
Two-source kernel: quadratic resolution, global error and full-profile injectivity	Exact for the model	Positive $W_{1/4}$ weight, Taylor remainder and Fourier uniqueness.
Spatial filter and cutoff derivative	Exact	Finite Euler sum, compensator, and full chain rule.
Quadratic pair inverse from finite samples	Conditional on a model residual	Linear independence, orthogonal projection, and explicit error estimate.
Separation of distant indices	Unconditional	$N_0 \pm H$ barriers and sublevel measure.
Local arithmetic occupation criterion	Exact	Saturated parts of the weighted prime phase.
Evaluation defects e_k and repeated heights	Exact	Consecutive index arithmetic and convexity.
2,001,052-zero computations	Numerical only	Repository text outputs; no independent interval certification.
Conditional $j = 1$ simple-critical-line fraction	Exact conditional inequality	A_T, B_T, fourth moment, derivative moment and boundary control.
Uniform estimate for all $X \geq T^{\varepsilon}$	Open	Requires additional arithmetic cancellation.
DBSRH⇔RH	Open	Requires a uniform explicit formula and non-cancellation, and careful handling of repeated imaginary parts.

The RvM-relative identity (21.3) expresses adjacent spacings through correlated prime displacements; Proposition 21.3 shows that the actual fixed truncation gives the coefficient $2\pi \cdot 0.08382$ rather than $2\pi C_{\mathrm{H}}$ for individual RvM deviations. Neither control establishes noncollapse. The author's MCOM estimate removes the truncation-scale defect of the original construction, and the Selberg–Fujii–Montgomery–Vaughan argument supplies a genuine, nontrivial uniform range of prime cutoffs. The structural objection is valid only at the level of the properties it explicitly assumes; it cannot replace analysis of the actual Euler phase. The unconditional local quadratic kernel theorem does not yet bound the residual from the actual Euler phase. The interval $X \geq T^{\varepsilon}$ without an upper restriction is the appropriate remaining *problem*, not an established theorem. The results here do not prove RH; neither the correlated-shift criterion nor the local inverse bounds have been established for every cutoff and every index.

Source note

The IJNT manuscript and the MCOM submission are used as cited inputs and are not modified here. The publicly accessible route notes and the saved ordinal-transport text from commit `38aac7c` are distinguished from the unpublished manuscripts. The new normal-family criterion is a mathematical derivation in this revision, not a result represented as already proved by the external notes. The numerical tables were cross-checked against the repository text outputs listed in [15, 16].

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Fujii–Selberg input and fourth moment of the untruncated phase. *(Cited on pp. 5, 6, 7, and 19.)*

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[3] V. Casaña-Giner, *Deep Blue Sea Riemann Hypothesis: convergence of prime-corrected sublevel points and the Riemann hypothesis*, research draft `teorema_DBSRH_v1.tex`, 2 November 2026; proposed weighted explicit formula and partial converse, with analytic steps still indicated as sketches. *(Cited on pp. 10 and 15.)*

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`mv_dphi_th0.5_j19.txt`, `tight_pairs_v13.txt`, `margin_triangle_v14.txt`; repository commit `2b24f67dc5348788714bafaf4a4c566aa041a877`. *(Cited on pp. 20 and 45.)*

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